

Name: _____

AS Physics Unit 1

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2016					
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	Max UMS
2016	27	34	41	48	55	69
2017	22	29	36	43	51	67
2018	27	34	41	48	56	72
2019	25	32	39	46	54	70
2022	10	17	24	32	40	56
2023	15	23	31	39	47	63

Surname	Centre Number	Candidate Number
Other Names		2



GCE AS/A Level
2420U10-1 – **NEW AS**



PHYSICS – Unit 1
Motion, Energy and Matter

A.M. TUESDAY, 24 May 2016
1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	10	
3.	13	
4.	12	
5.	11	
6.	12	
7.	14	
Total	80	

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use pencil or gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the continuation pages at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The total number of marks available for this paper is 80.
The number of marks is given in brackets at the end of each question or part-question.
You are reminded to show all working. Credit is given for correct working even when the final answer given is incorrect.
The assessment of the quality of extended response (QER) will take place in **Q5(a)**.



MAY162420U10101

Answer all questions.

1. (a) A student wishes to determine the material from which a metal ball bearing is made. He obtains the following values and uses them to determine the density of the metal in the ball bearing.

Volume of ball bearing = $5.6 \pm 0.2 \text{ cm}^3$
Mass of ball bearing = $45.4 \pm 0.5 \text{ g}$

- (i) Calculate the density of the ball bearing (in g cm^{-3}) and show that its **percentage** uncertainty is approximately 5%. [3]

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- (ii) Determine the **absolute** uncertainty in the density. [1]

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- (b) The table gives the density of some common metals and alloys.

Metal	Density / g cm^{-3}
Tin	7.3
Stainless steel	7.5
Iron	7.9
Brass	8.2
Nichrome	8.3
Copper	8.9



Examiner
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- (i) Write down possible materials from which the ball bearing is made and explain why it is not possible to determine the exact material. [2]

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- (ii) Identifying the correct material from which the ball bearing is made depends on reducing the uncertainty in density. Explain which of the two values (volume or mass) contributes more to this uncertainty. [2]

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2. (a) State the difference between *baryons* and *mesons* in terms of quark make-up. [2]

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(b) When two protons collide, the following interaction may occur, where *x* is an unknown particle:

$$p + p \longrightarrow p + x + \pi^0$$

(i) The π^0 is a meson which carries no charge. State its quark make-up. [1]

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(ii) Identify particle *x*, explaining how you use the law of conservation of baryon number and one other conservation law. [3]

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(iii) State how lepton number is conserved in the above interaction. [1]

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Examiner
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(c) A π^0 decays in a typical time of 8×10^{-17} s into two photons as shown. State which force is involved in this interaction, giving **two** reasons for your answer. [3]



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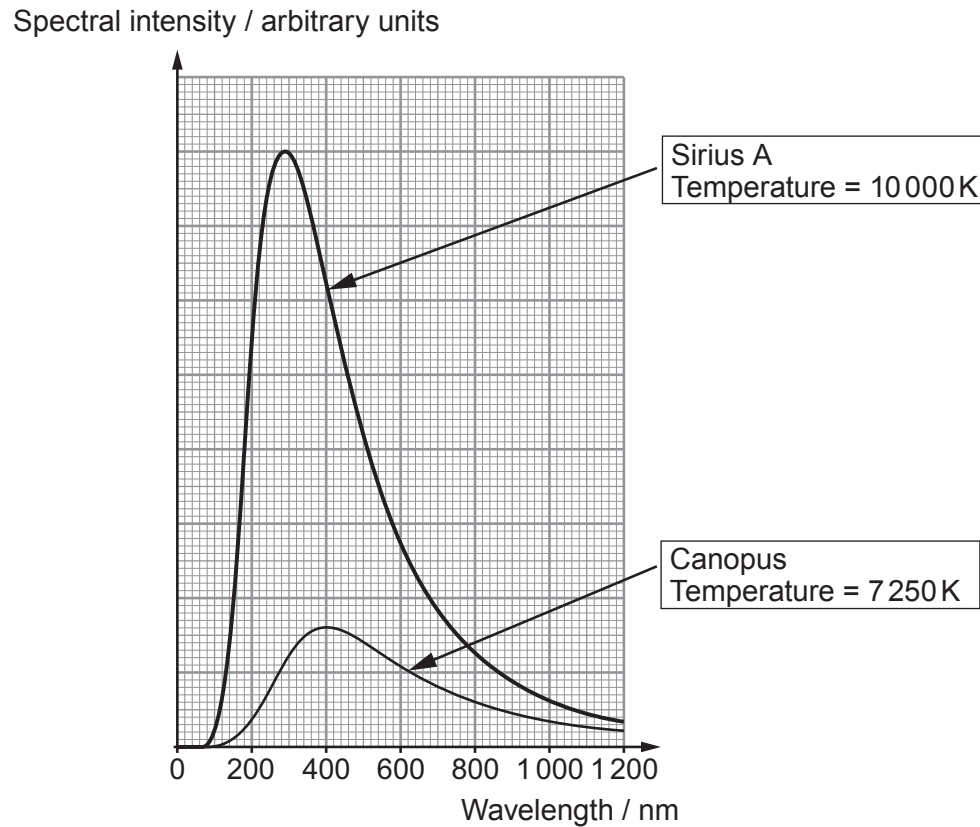
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3. (a) Two of the brightest stars in the night sky are Sirius A and Canopus. The graph shows the continuous black body spectra for these two stars.



- (i) Confirm that Wien's displacement law is valid for the stars. [3]

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(ii) Astronomers state that one of the stars appears ‘bluer’ than the other. Explain how the spectra support this statement and state which star would appear bluer. [2]

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(b) (i) The radius of Canopus is 4.97×10^{10} m and that of Sirius A is 1.19×10^9 m. Show that the luminosity of Canopus is approximately 500 times the luminosity of Sirius A. [4]

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(ii) Calculate the intensity of the radiation reaching the surface of the Earth from Sirius A. (Distance between Sirius A and Earth = 8.15×10^{16} m.) [2]

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(iii) The intensity of the radiation reaching the Earth’s surface from Canopus is **less than** that from Sirius A, even though Canopus has a **greater** luminosity than Sirius A. Explain this apparent contradiction. [2]

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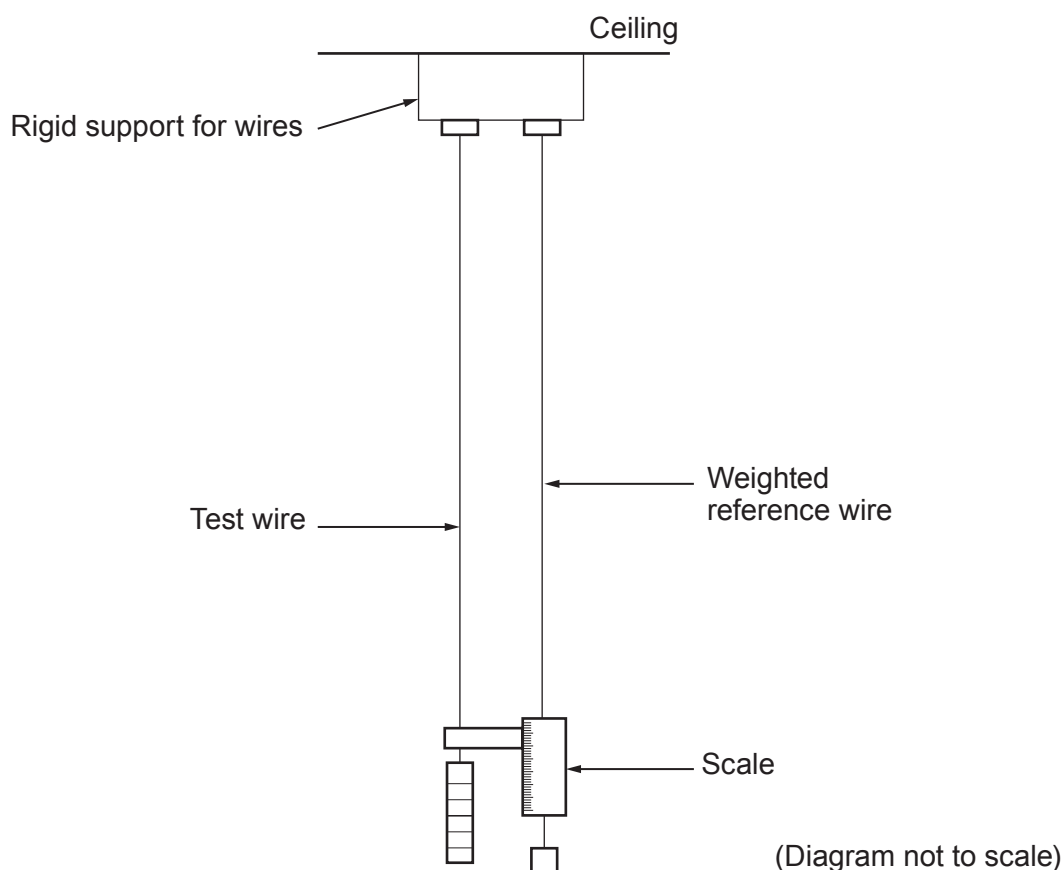
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4. The following extract from a Physics text book describes a method for determining the Young modulus of a metal in the form of a wire.

Apparatus

A long test wire and a reference wire of the same length and material are suspended from a common rigid support. This minimises the effect of temperature and the movement of the support. The scale for measuring extensions is on the reference wire and a weight is placed on it to keep it taut and kink free. This way if the test wire pulls the support downwards, the reference wire and scale move with it. The scale will therefore read only the extension of the test wire.



Method

Measure the extension for increasing loads ensuring that the wire remains within its *elastic limit*. Take repeat readings during the removal of the load to provide a mean value for the extension. Measure the diameter of the test wire at several different places using a micrometer. Measure the length of the test wire with a ruler.

- (a) (i) Explain how the effect of a change in temperature on the wire is minimised. [2]

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(ii) State what is meant by the term *elastic limit* **and** explain how an experimenter would know whether or not the test wire has extended beyond its elastic limit. [2]

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(iii) The text book also states:

‘The apparatus and procedures are designed carefully to minimise uncertainties when taking measurements.’

I. State why a long wire is used rather than a short one. [1]

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II. Why is the diameter of the wire measured at several different places? [1]

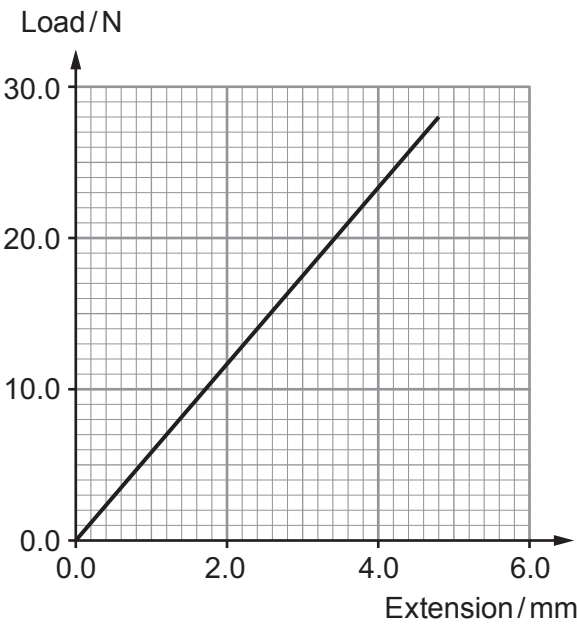
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(b) A typical graph showing the results of such an experiment is given below. The original length of the wire is 2.40 m and the mean value of its **diameter** is 0.32 mm.



(i) Use the graph and the measurements given to determine the Young modulus of the material of the wire. Give your answer to an appropriate number of significant figures. [4]

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(ii) Calculate the energy stored in the wire when it is extended by 2.4 mm. [2]

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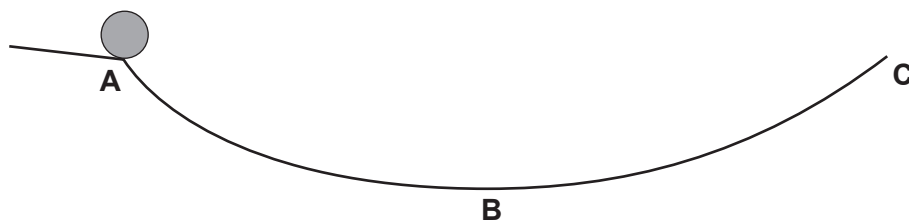
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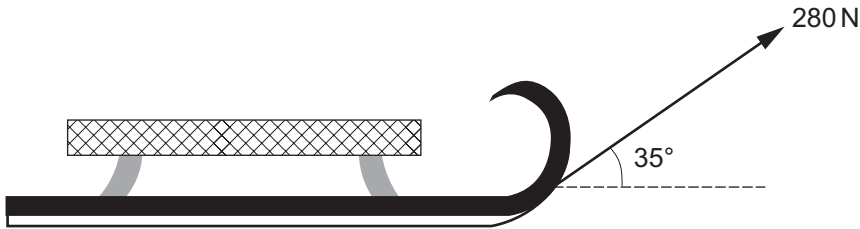
5. (a) The diagram shows a marble about to be released at **A** on a curved ramp.



Explain the energy transfers that take place from the moment the marble is released to the moment it finally comes to rest. [6 QER]



- (b) A heavy sled moves with **constant velocity** when it is pulled by a force of 280 N acting as shown.



- (i) State why the work done in pulling the sled **cannot** be calculated by simply multiplying 280 N by the distance the sled is pulled. [1]

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- (ii) It takes 20 minutes to pull the sled a distance of 3.0 km over level ground. Calculate the mean power needed. [4]

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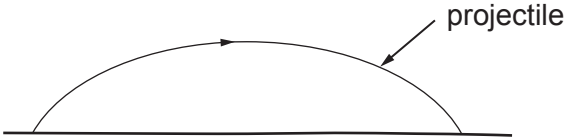
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6. (a) A projectile is fired from the Earth's surface and follows a curved path as shown.



Describe and explain how, if at all, the vertical and the horizontal components of velocity of the projectile change during the flight. Ignore the effects of air resistance. [3]

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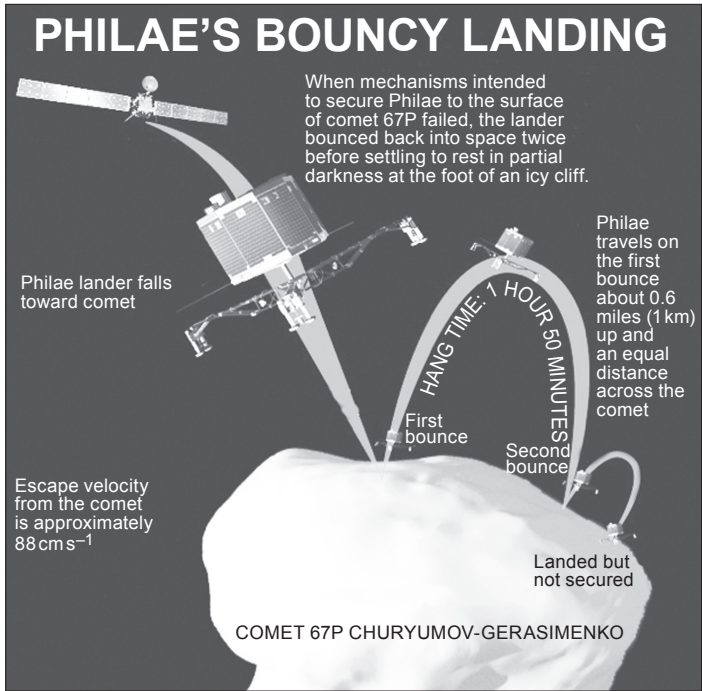
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(b) In November 2014 the space probe ‘Philae’ was dropped onto comet 67P. Philae bounced twice before coming to rest. A national newspaper used the following image to describe the landing.



(i) Use information from the image to calculate the horizontal velocity of the lander between the first and second bounce. [1]

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(ii) By considering the first bounce, show that the value of the acceleration due to gravity on the comet is approximately 0.0002 ms^{-2} . [2]

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(iii) Show that the vertical velocity of the lander immediately after the first bounce was greater than 60% of the escape velocity. [3]

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(c) By considering the following facts discuss whether or not the mission was justified. [3]

- Cost of developing and sending the spacecraft to the comet: £1 billion over 10 years.
 - Around 2 000 people involved in the development of the spacecraft and its instruments.
 - Advanced solar cell technology developed.
 - 28 000 landing announcement ‘re-tweets’ in the first hour.
 - Organic molecules detected on comet surface.

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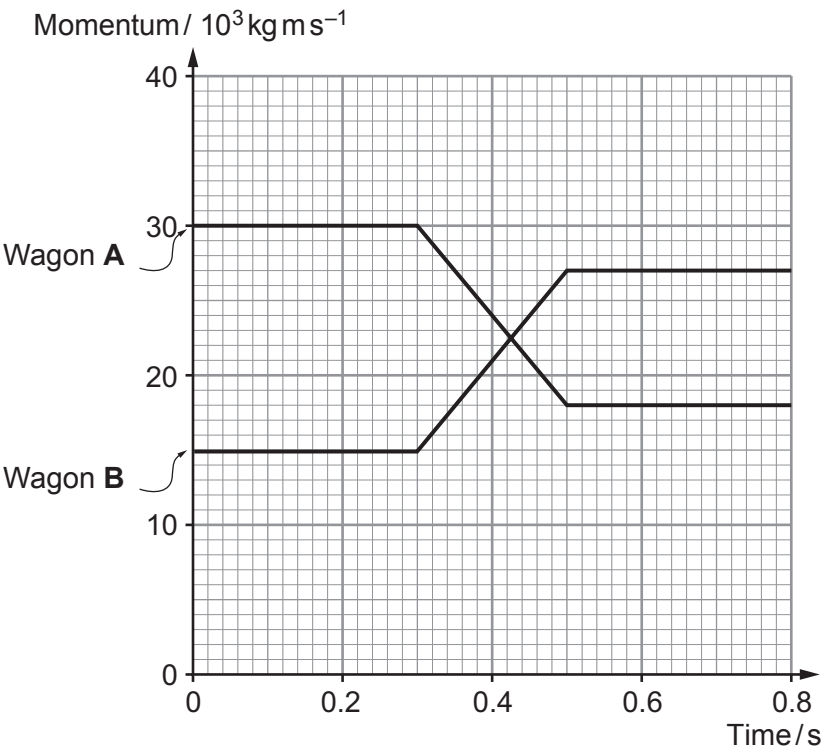


7. (a) Momentum is a *vector*. State what is meant by a vector. [1]

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(b) The graph shows how the momentum of two colliding railway wagons (**A** and **B**) varies with time. The collision takes place between 0.30 s and 0.50 s as shown. The wagons remain joined together after impact.



(i) Explain with suitable calculations how the graphs show that external forces on the system are negligible. [3]

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Examiner
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(ii) Calculate the final velocity of the two wagons given that the total mass of wagon **A** and wagon **B** is 25 000 kg. [3]

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(c) (i) State Newton’s Second Law of motion in terms of momentum. [2]

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(ii) Determine from the graph the resultant force on wagon **A** during the collision. [3]

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(iii) The force experienced by wagon **B** during the collision is equal and opposite to the force experienced by wagon **A**. State which law of motion this is an example of **and** explain how the graph confirms this law. [2]

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END OF PAPER





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GCE AS/A LEVEL – **NEW**

2420U10-1



PHYSICS – AS unit 1
Motion, Energy and Matter

TUESDAY, 23 MAY 2017 – MORNING

1 hour 30 minutes

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The assessment of the quality of extended response (QER) will take place in **Q7**.



MAY172420U10101

Answer all questions.

1. (a) In the production of steel alloy, atoms of carbon are added to iron. The resulting alloy is less *ductile* than pure iron. State the meaning of the term *ductile*, and describe, on an atomic scale, why the addition of carbon atoms can make steel less ductile than iron. [3]

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- (b) A wire of length 2.4m and diameter 0.60mm is made of steel of Young modulus $200 \times 10^9 \text{Nm}^{-2}$. The wire is loaded so that its length is increased by 1.8mm. Assuming that the change is elastic, calculate:

(i) the strain; [1]

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(ii) the applied stress; [2]

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(iii) the force applied to the wire; [2]

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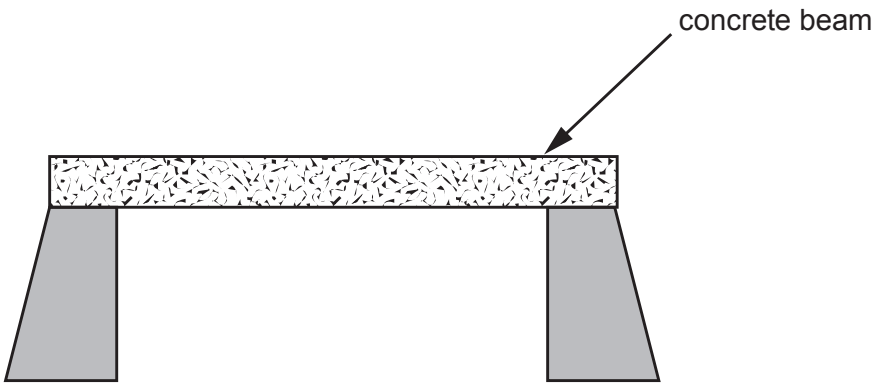
(iv) the elastic energy stored in the wire. [2]

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(c) The diagram shows how a gap can be bridged using a concrete beam.



(i) **On the diagram** label a point in tension with the letter **T**, and a point in compression with the letter **C**. [1]

(ii) Inserting a pre-stressed steel bar into the concrete beam would increase the breaking stress of the concrete. **On the diagram**, draw a pre-stressed steel bar in an appropriate position. [1]

(iii) Explain how the steel bar strengthens the beam. [2]

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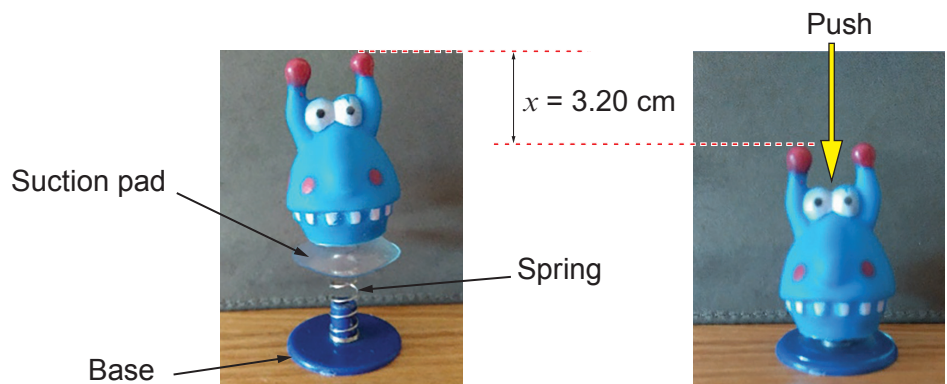
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2. A student is asked to find a value for the spring constant, k , of a spring used in a jumping toy. Pushing down on the toy compresses the spring allowing the suction pad to stick to the base. After a few seconds the force on the suction pad decreases and the toy ‘jumps’ into the air.



The student measures the jump height for five separate jumps using a metre rule. The results are shown in the table.

Jump number	1	2	3	4	5
Jump height / cm	48	52	54	46	49

- (a) (i) Calculate the mean jump height, h_{mean} along with the **absolute** uncertainty in its value. [2]

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- (ii) Calculate the **percentage** uncertainty in h_{mean} . [1]

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- (b) (i) The student applies a principle of physics to the jumping toy to show that:

$$k = \frac{2mgh_{\text{mean}}}{x^2}$$

where: m is the mass of the toy and x is the compression of the spring.

State which principle was used and explain how it can be applied to derive the formula shown. [2]

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(ii) The student uses an accurate balance to measure the mass, m , of the toy to be 48.40 g and digital callipers to measure the compression of the spring, x , to be 3.20 cm. He decides **not** to determine the uncertainty in these measurements.

I. Explain why it is reasonable for the student to ignore these uncertainties. [2]

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II. Calculate k along with the **absolute** uncertainty in its value. Give both values to an appropriate number of significant figures. [3]

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III. State whether your answer to (b)(ii)(II) is likely to be smaller than or greater than the actual value for k . Justify your answer. [2]

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(c) Describe one practical procedure by which the student could reduce the uncertainty in k . [1]

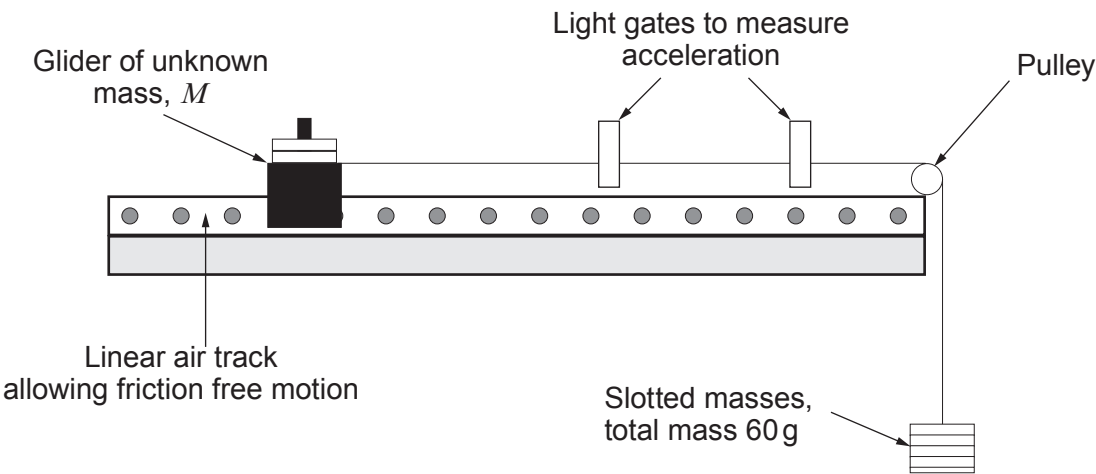
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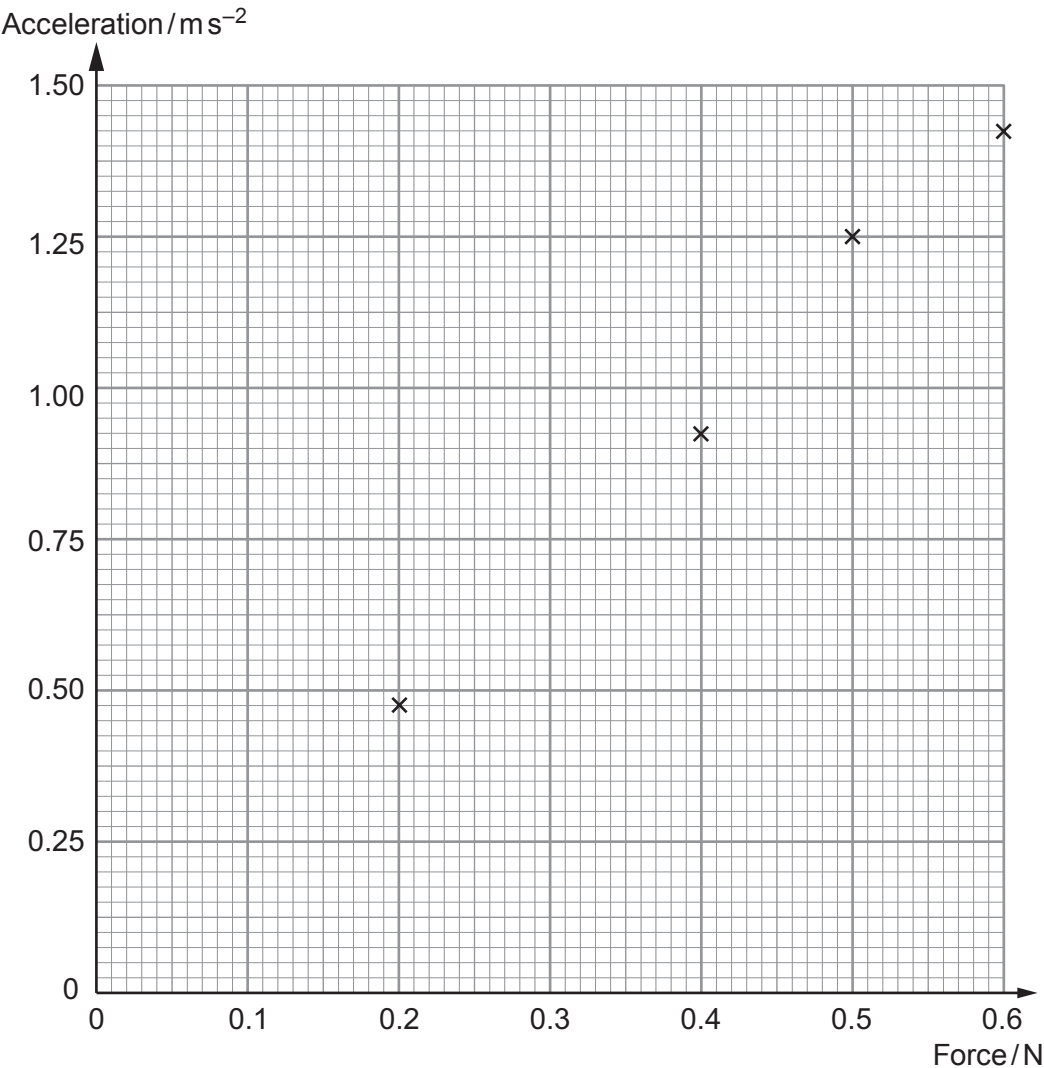


3. (a) A student uses the following apparatus to determine the unknown mass, M , of a glider.



The total mass of the system is kept constant by removing one of the slotted masses from the hanging weight and placing it on the glider for each reading of force and acceleration.

She plots **some** of her measurements on the grid below.



Examiner
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(i) Draw a line of best fit on the graph and use it to determine the gradient. [2]

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(ii) Show clearly that the gradient has units kg^{-1} . [2]

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(iii) Determine the value of M , the unknown mass of the glider. [3]

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(iv) Comment on the quality and sufficiency of the data obtained. [2]

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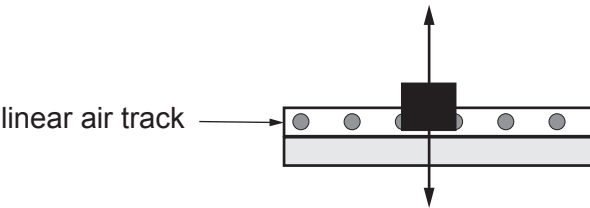
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(b) Two vertical forces acting on the glider are shown below.

Force on glider due to air =N



Weight of glider =N

- (i) **Fill in the missing values** on the above diagram. [2]
- (ii) The two forces shown **do not** form a Newton 3rd law pair. Give two reasons why. [2]

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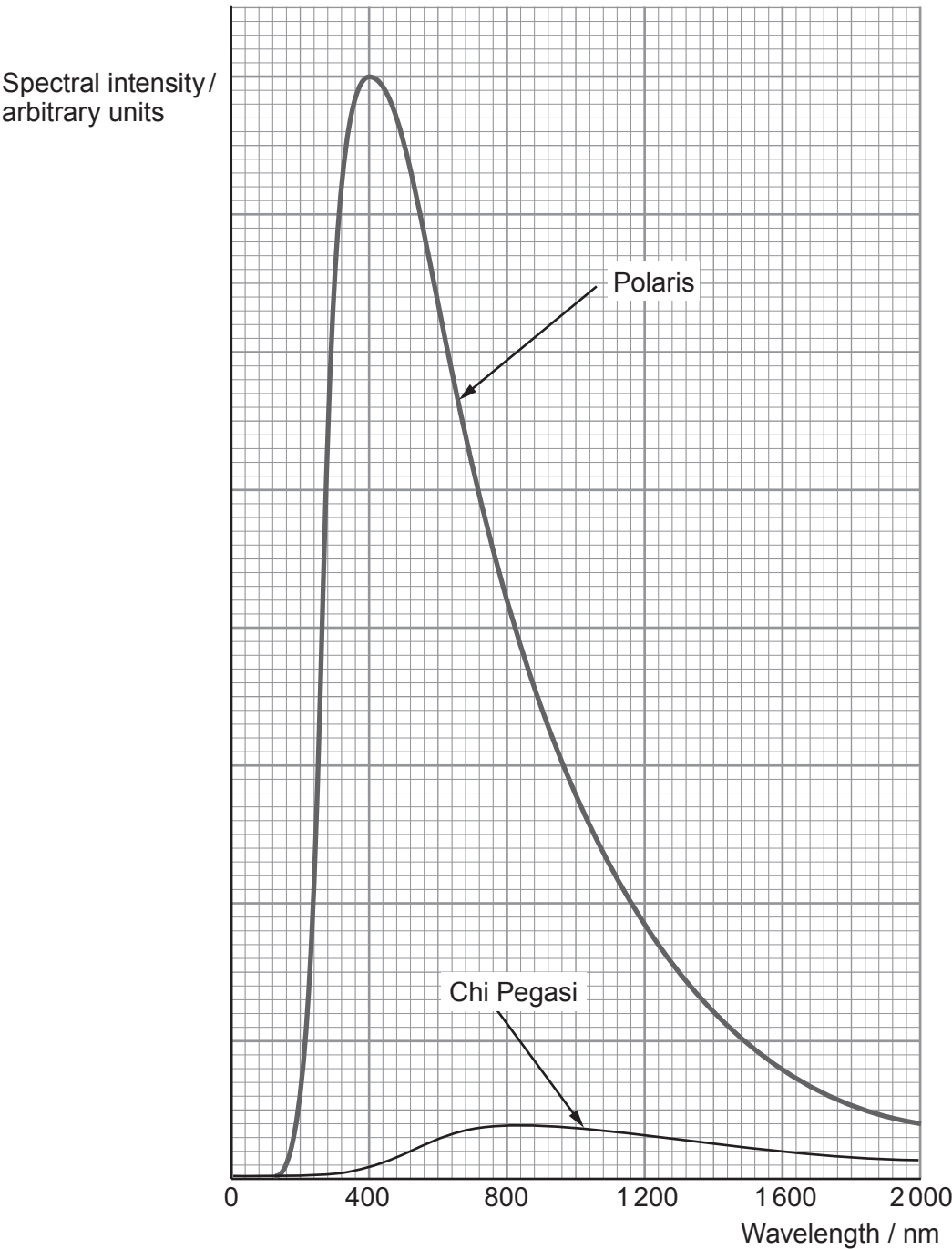
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4. (a) Stars are very good approximations to *black bodies*. State what is meant by a *black body*. [1]

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(b) The graph shows the black body radiation curves for the two stars Polaris (sometimes called the North Star) and Chi Pegasi (a red supergiant in the constellation Pegasus). The stars are equidistant from the Earth.



Examiner
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(i) Use the graph to state **three** differences between Polaris and Chi Pegasi. [3]

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(ii) The surface temperature of Polaris is 7 250 K. How does the graph confirm this? [2]

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(iii) Polaris is 431 light years from Earth and the intensity of radiation received on Earth from it is $4.05 \times 10^{-9} \text{ W m}^{-2}$. Show that the luminosity of Polaris is approximately $8.5 \times 10^{29} \text{ W}$. [1 light year = $9.46 \times 10^{15} \text{ m}$] [2]

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(iv) Calculate the radius of Polaris. [3]

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5. (a) Two equations of accelerated motion are $v = u + at$ and $v^2 = u^2 + 2ax$. Use these equations to show that, for a body accelerating uniformly from rest: [2]

$$x = \frac{1}{2} at^2$$

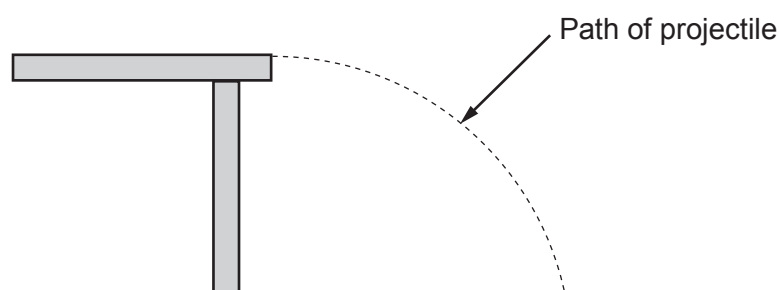
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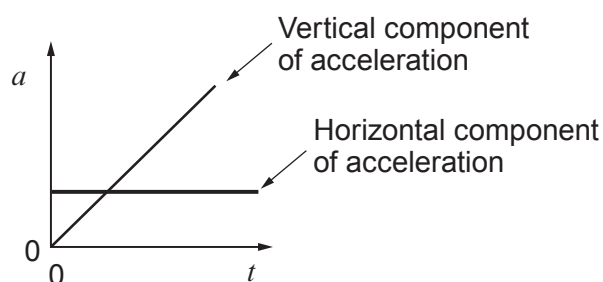
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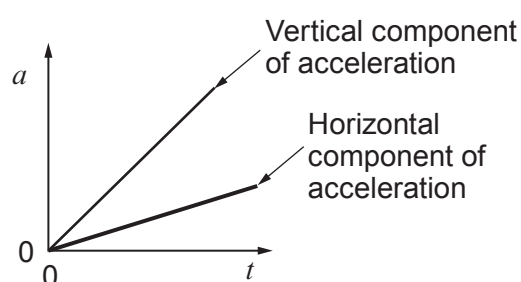
- (b) The diagram shows the path of a projectile after it is launched horizontally from a table.



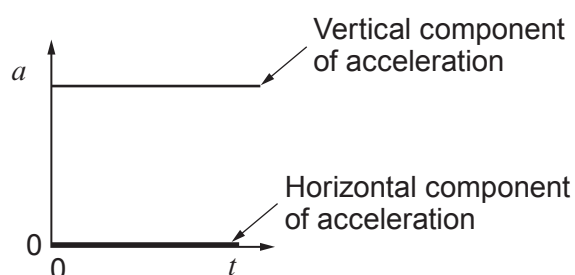
When considering the horizontal and vertical components of the motion, graphs of horizontal acceleration and vertical acceleration against time are sketched. **Only one** of the following sketch graphs shows a correct combination. *[Ignore air resistance for the remainder of the question.]*



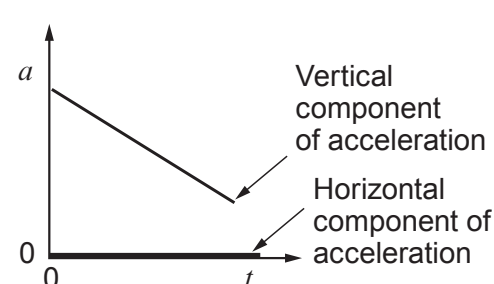
Graph 1



Graph 2



Graph 3



Graph 4



State which graph shows the correct combination and explain your answer. [3]

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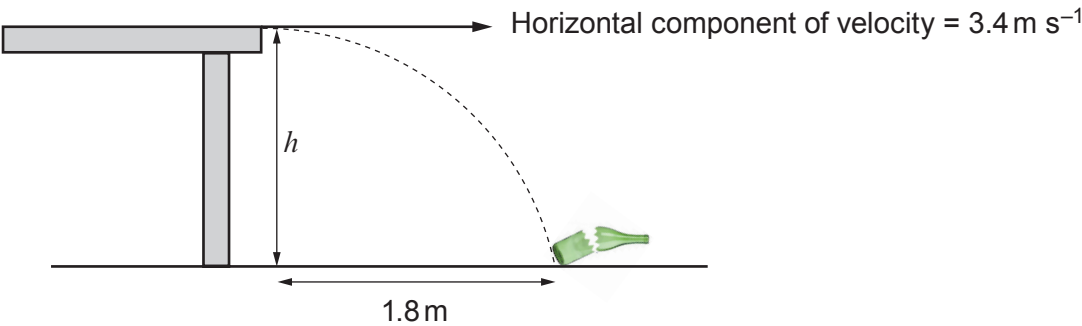
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(c) A bottle is accidentally knocked from the table and follows the path shown.



(i) Calculate the height, h , of the table. [3]

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Examiner
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(ii) Calculate the magnitude of the velocity and the direction of travel of the bottle just before it hits the ground. [4]

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(d) State whether or not the following statement is correct and justify your answer. [2]

The flight time for the bottle in part (c) will depend on the horizontal velocity – the greater the horizontal velocity, the longer it will take for the bottle to hit the floor after leaving the table.

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6. (a) State the conditions necessary for a body to remain in equilibrium.

[2]

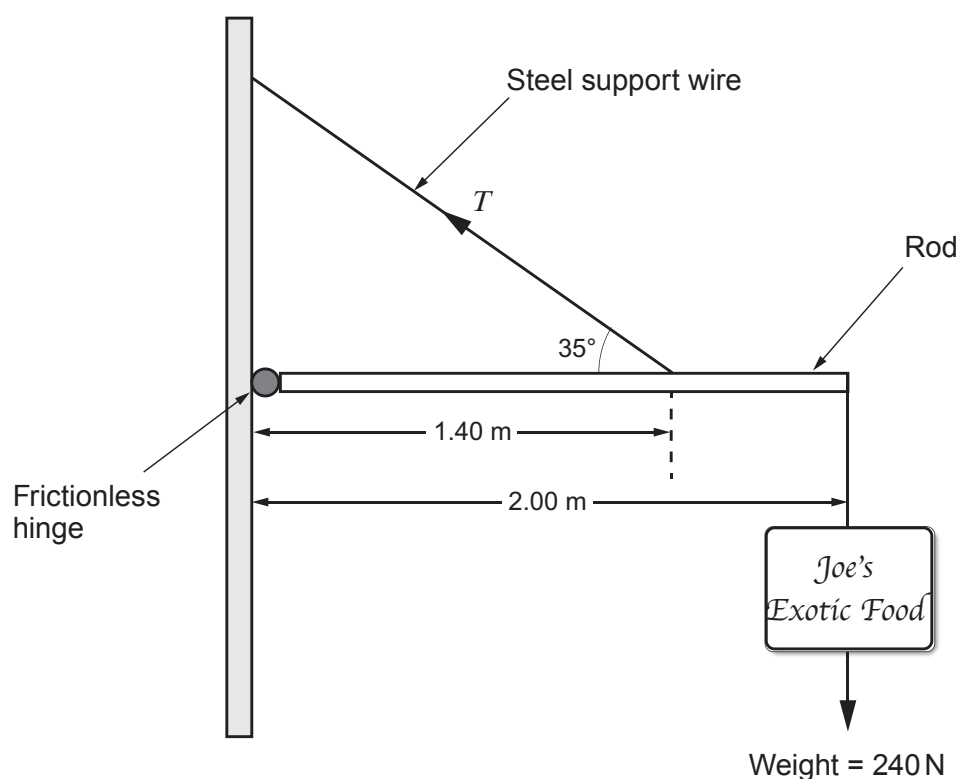
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- (b) A shopkeeper wishes to hang a heavy sign of weight 240 N outside his shop using the following arrangement. He has steel wires of various diameters to choose from. You may neglect the weight of the rod.



The shopkeeper needs to ensure that the steel wire he decides to use has a breaking strength greater than is required to support the sign. By taking moments about an appropriate point, show that the tension, T , in the wire in the above diagram is approximately 600 N.

[3]

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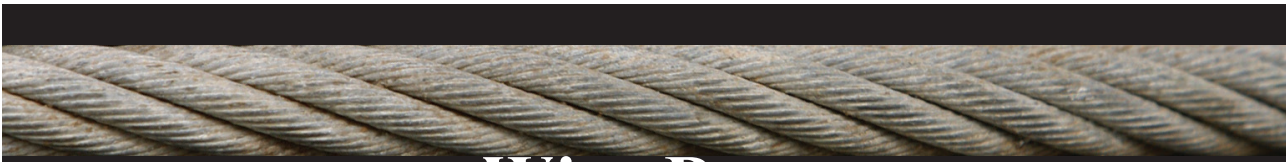
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(c) The shopkeeper finds the following information on a website advertising steel wire ropes.



Wire Rope

WARNING

Failure to read, understand, and follow manufacturer instructions may cause death or serious injury.

Minimum breaking strength and safe working load of uncoated steel wire ropes are indicated below.

Diameter of rope /mm	Minimum breaking strength /N	Safe working load /N at 5:1 ratio	Safe working load /N at 3:1 ratio
1.5	1 900	380	633
2.0	2 750	550	916
2.5	3 300	660	1 100
3.0	5 400	1 080	1 800

Loading Information

SAFE WORKING LOAD (SWL) is the load that can be applied without causing any damage to the wire rope.

WE PROVIDE TWO SWLs FOR YOUR CONSIDERATION AT RATIOS 5:1 AND 3:1.

Factors of safety should always be applied when determining maximum wire rope loading conditions. If in doubt a suitably qualified engineer should be consulted to assess loading factors.

 Search the web and windows





Examiner
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- (i) Based on the information on the web page the shopkeeper decides to apply a SWL ratio of 3:1 for the wire he will use. State, giving your reasoning, which **minimum** diameter of wire rope he should choose to use. [2]

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- (ii) The shopkeeper has no engineering background. Evaluate whether or not he has made an informed decision. [2]

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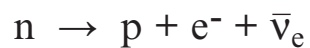
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**TURN OVER FOR THE
LAST QUESTION**



Examiner
only

7. The following process describes the decay of a neutron.



Give a detailed description of the process, including how conservation laws apply. [6 QER]

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

END OF PAPER

6



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AS/A LEVEL
2420U10-1
PHYSICS – AS unit 1
Motion, Energy and Matter



TUESDAY, 15 MAY 2018 – MORNING
1 hour 30 minutes

For Examiner's use only		
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Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the continuation page at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The total number of marks available for this paper is 80.
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **6(a)**.



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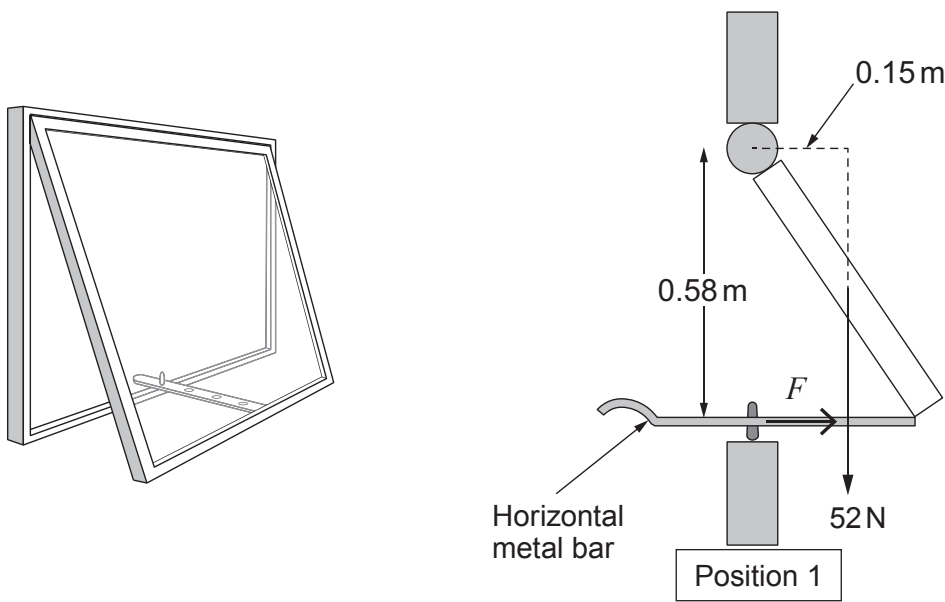
Answer all questions.

1. (a) State in words the equation used to calculate the moment of a force about a point. [1]

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(b) The picture and diagram show a window hinged at the upper surface. The window is opened by pushing on the horizontal metal bar attached to its lower surface. Holes are drilled into the metal bar so that the window can be supported at various opening positions, one of which is shown below and labelled as Position 1. The hinge provides no resistance to the movement of the window.



(i) Show that the clockwise moment produced by the weight of the window is approximately 8 N m. [1]

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(ii) Hence calculate the force, F , the metal bar exerts on the window. [2]

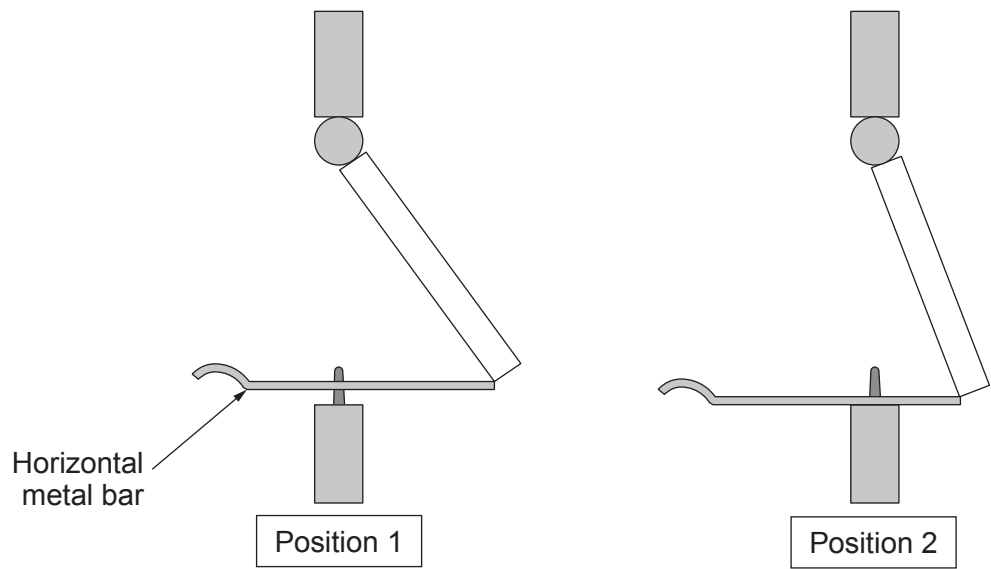
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(c) Tom and Bethan discuss how the force in the metal bar changes with changing positions. Tom thinks that the force in the bar is greater when the window is in Position 2, whereas Bethan believes that the force is greater when the window is in Position 1. Discuss who is correct, giving a detailed explanation in terms of moments. Assume the metal bar is horizontal in both positions. [4]



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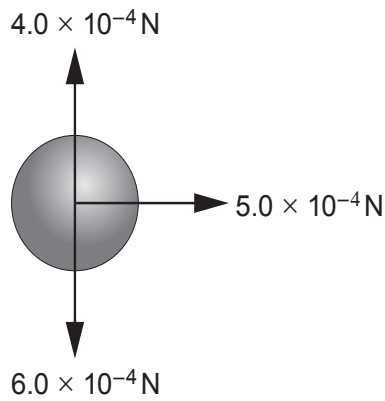
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2. (a) The forces acting on a hailstone falling in a horizontal cross-wind can be represented as in the diagram.



- (i) Calculate the magnitude and direction of the resultant force acting on the hailstone. [3]

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- (ii) At a later time, the wind has stopped blowing and the hailstone falls at terminal velocity. In terms of forces, explain why the hailstone is at terminal velocity. [1]

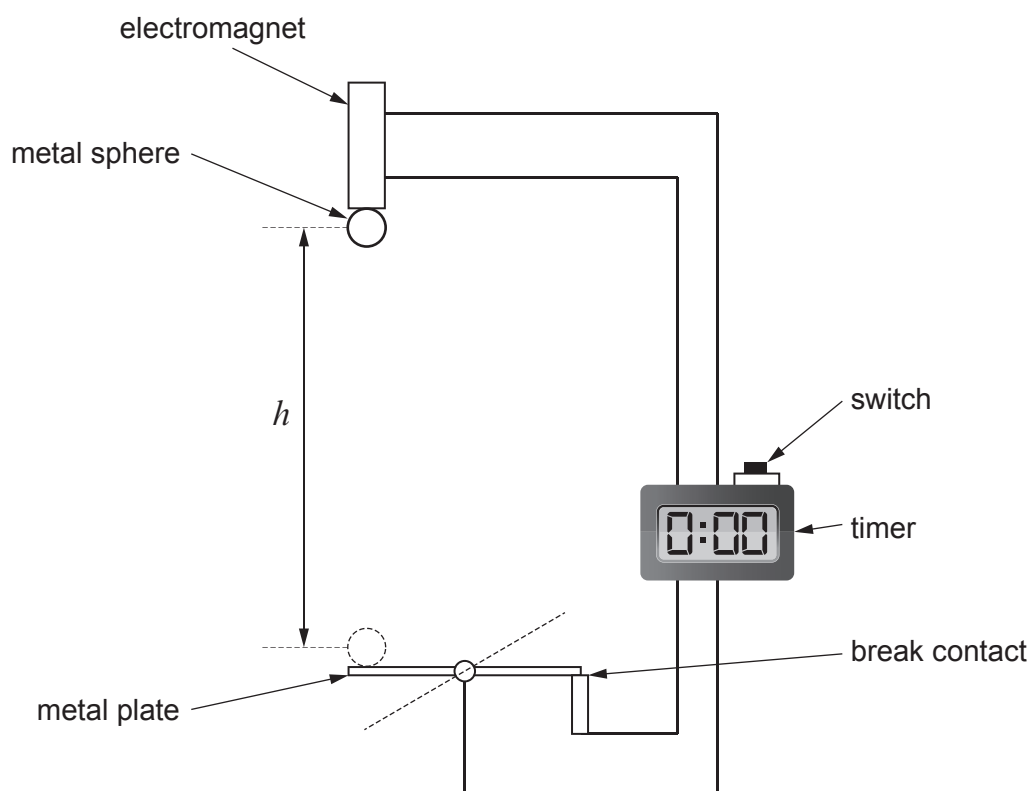
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- (b) Aled uses the following apparatus to measure the acceleration of free-fall, g .



When the switch is pressed, it starts the timer and disconnects the electromagnet, almost instantly releasing the metal sphere. When the sphere hits the metal plate it breaks the circuit, stopping the timer. The time taken for the metal sphere to fall through a range of different heights, h is measured.

- (i) Aled is told that there is a very small time delay between the switch being pressed and the ball being released. This is a systematic error. The manufacturer states that the time delay is 0.05 s. State how Aled should account for the systematic error when taking readings. [1]

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(ii) Aled records his **corrected results (i.e. with the systematic error accounted for)** in the table below. Complete the row for time squared, t^2 giving your answers to an appropriate number of significant figures. [2]

Drop height, h/m	0.40	0.80	1.20	1.60	2.00
Corrected time, t/s	0.27	0.41	0.48	0.58	0.64
Corrected time squared, t^2/s^2					

(iii) The following relationship is used to find a value for g :

$$g = \frac{2h}{t^2}$$

Show how this relationship is obtained from an appropriate equation of accelerated motion. [2]

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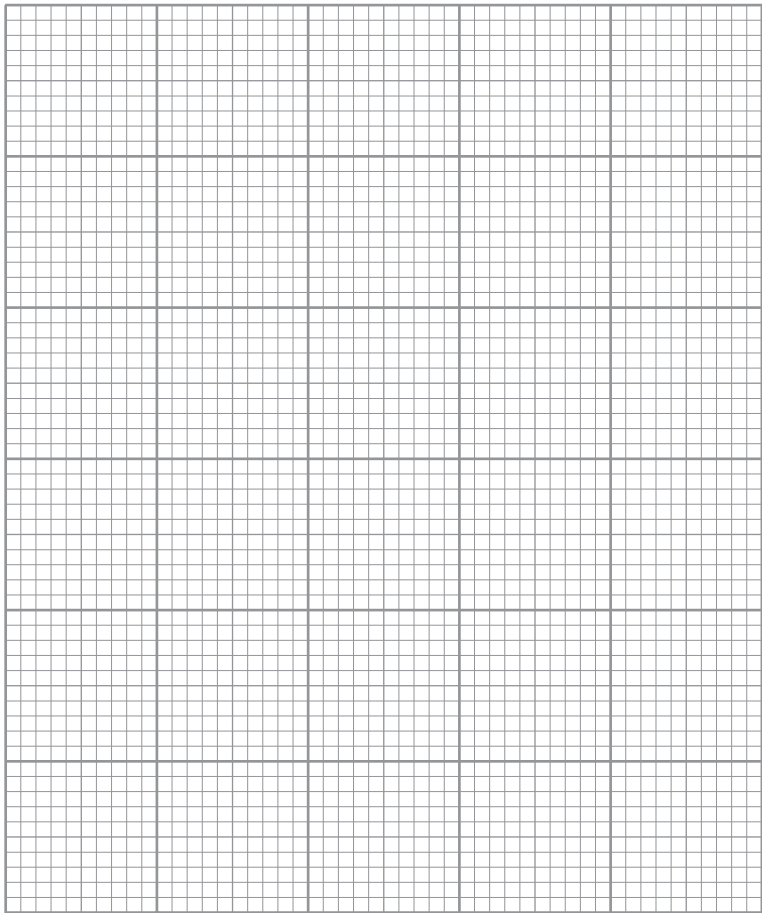
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Examiner
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- (iv) On the grid below, plot a graph of h (**vertical axis**) against t^2 (**horizontal axis**) and draw a line of best fit. [4]



- (v) Use your graph to determine a value for g . [3]

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- (c) Discuss to what extent your graph agrees with the equation in (b)(iii). [3]

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3. (a) Describe a method to investigate the force-extension properties of rubber in the form of an elastic band **as it is loaded**. You should describe how the extension of the rubber is accurately measured. [3]

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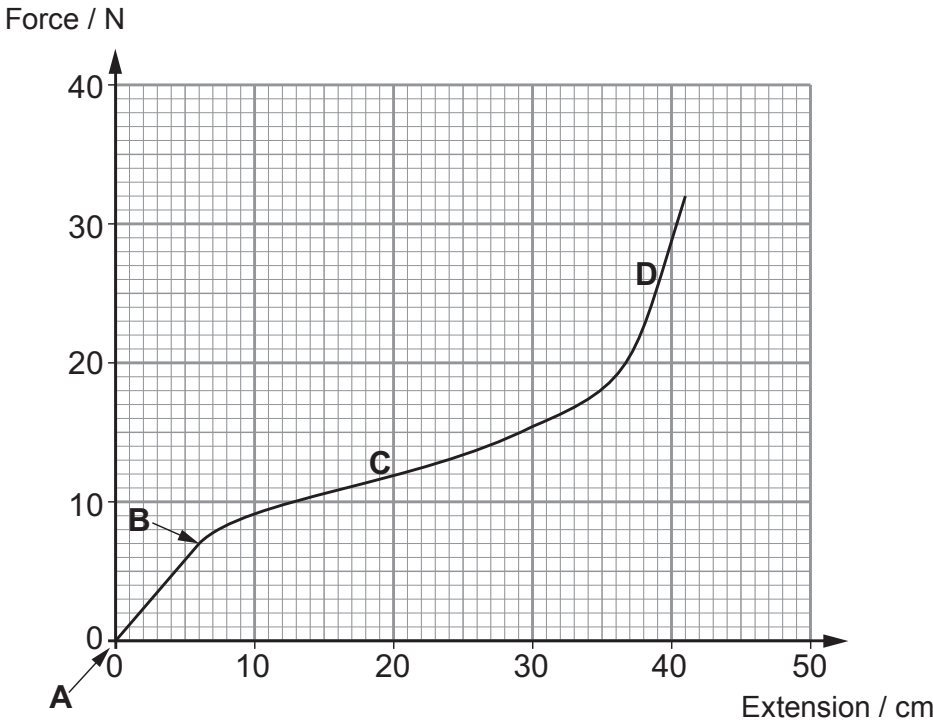
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- (b) The results from such an experiment for a rubber band of unstretched length 8.0 cm are plotted in a graph.



Examiner
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(i) Calculate the strain in the rubber at point **B**. [1]

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(ii) Determine the Young modulus of the rubber in the region **AB**. Assume the band has a total cross-sectional area of 0.050 cm^2 . [3]

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(c) By referring to the molecular structure of rubber, explain why the gradient at **C** is less than the gradient at **D**. [3]

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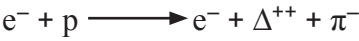
4. (a) The table shows information about some sub-atomic particles.

Particle	Symbol	Quark combination	Charge / e	Baryon number
proton	p	uud	+1	1
delta particle	Δ^{++}	uuu		
electron	e^{-}	no quarks present		
pion	π^{-}		-1	

(i) Complete the table. [3]

(ii) Identify the lepton in the table. [1]

(b) JJ Thomson, when studying the properties of cathode rays in 1897, discovered the electron. In the early 20th century, Ernest Rutherford, carrying out a series of experiments on radioactive substances, discovered the proton. The following interaction between protons and electrons has been observed by using high energy particle accelerators.



Show how charge and lepton number are conserved in the above interaction. [2]

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Examiner
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(c) The Δ^{++} decays in about 6×10^{-24} s as shown below.



(i) Show clearly that both up-quark number and down-quark number are conserved in this decay. [2]

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(ii) Give **two** reasons for believing that this decay is a strong force interaction. [2]

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(d) During a press conference, the spokesman for a nuclear research centre was asked the question:

‘You have discovered many new particles, none of which have had any discernible impact on society. How do you justify the huge expense of continuing with these experiments?’

In response, the spokesman referred to the work of JJ Thomson and Ernest Rutherford. Suggest why the spokesman responded in this way. [2]

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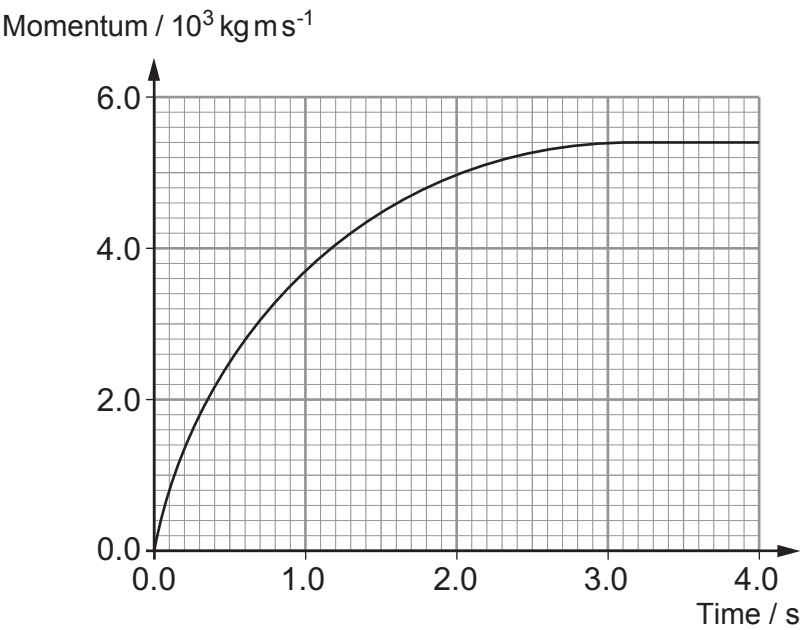


5. (a) A law of motion can be expressed as:

$$\text{resultant force} = \frac{\text{change in momentum}}{\text{time}}$$

State the name of the law. [1]

(b) The graph shows how the momentum of a spacecraft varies with time.



(i) By drawing a suitable tangent, show that the resultant force on the spacecraft at $t = 1.0 \text{ s}$ is approximately 2 kN . [2]

(ii) Hence show that the mass of the spacecraft is approximately 5000 kg , given that its acceleration at $t = 1.0 \text{ s}$ is 0.4 m s^{-2} . [1]

(iii) Label, with the letter **P**, a point on the graph where the resultant force on the spacecraft is zero. [1]



Examiner
only

(c) At $t = 4.0\text{ s}$ the spacecraft ‘docks’ (collides) with another **stationary** spacecraft of mass $7\,000\text{ kg}$. They join on impact.

(i) State the principle of conservation of momentum. [2]

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(ii) Calculate the velocity of both spacecraft after colliding. [3]

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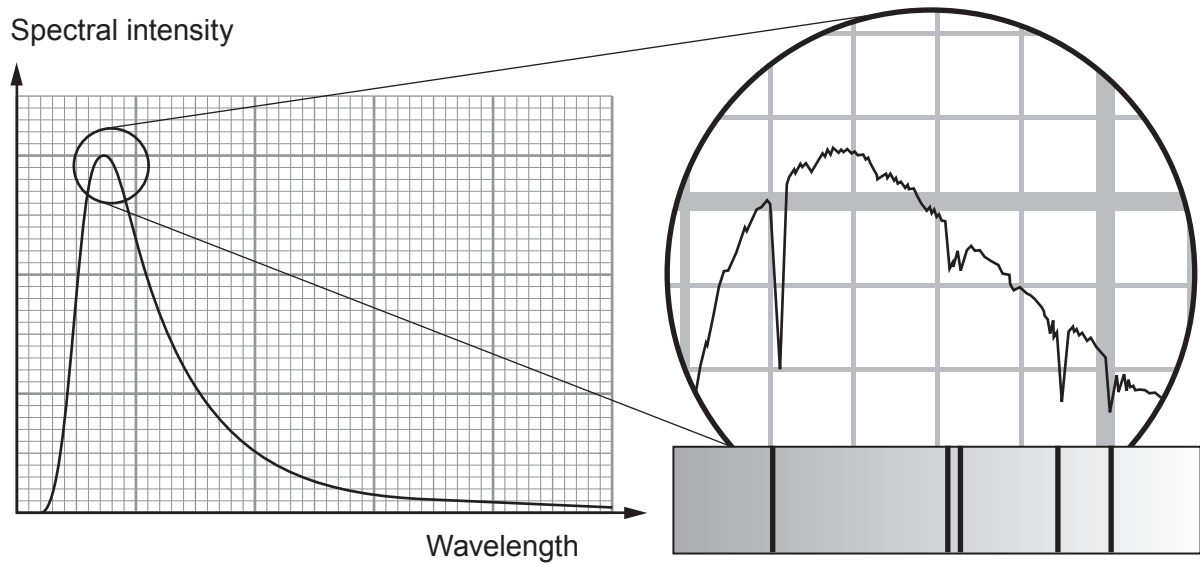
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Examiner
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6. (a) A blackbody graph of spectral intensity against wavelength for a star is shown. A magnified section, showing the finer detail of the spectrum is also given. An associated line spectrum is also shown.



Explain how the graph and the spectra can be used to provide information about the star and the elements from which it is made. [6 QER]



Examiner
only

- (b) (i) Altair is the brightest star in the Aquila constellation. It is 1.58×10^{17} m away, and the intensity of its electromagnetic radiation reaching the Earth is $1.32 \times 10^{-8} \text{ W m}^{-2}$. Show that its luminosity is approximately $4 \times 10^{27} \text{ W}$. [3]

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- (ii) Calculate Altair's **diameter** given that its surface temperature is 7 700 K. [3]

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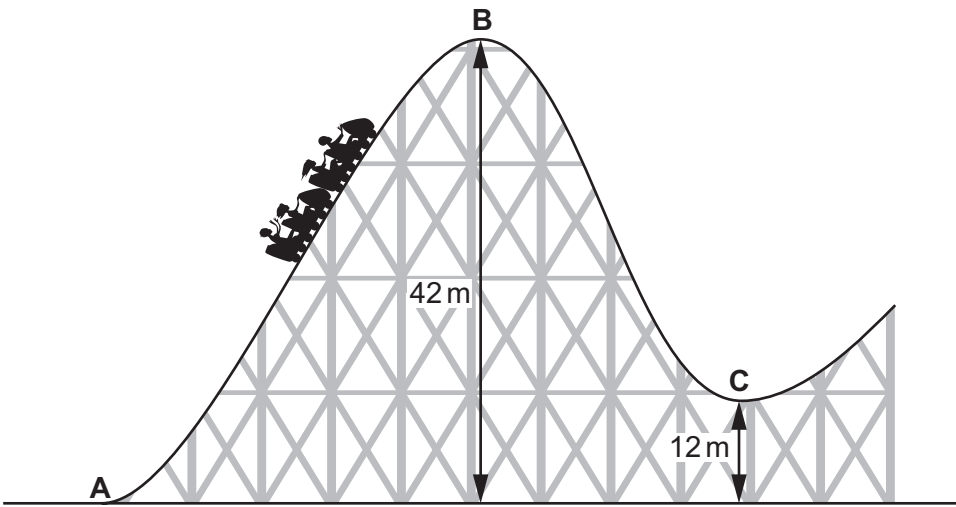
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TURN OVER FOR THE
LAST QUESTION



7. The diagram shows part of a rollercoaster ride at a theme park.



(a) A motor with a power output of 65 kW and a chain mechanism pulls the carriages of mass 2 600 kg from **A** to **B** in a time of 32 s.

(i) Show that the work done by the motor in 32 seconds is approximately 2 MJ. [1]

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(ii) Hence calculate the efficiency of the mechanism, assuming the carriages are momentarily at rest at **B**. [3]

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Examiner
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(b) At B, the carriages become disconnected from the motor and the carriages move under the influence of gravity for the rest of the ride. In moving from B to C, a distance along the track of 36 m, the carriages experience a mean resistive force of 2.8 kN. Calculate the speed of the carriages at C. [5]

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END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		2



AS/A LEVEL
2420U10-1
PHYSICS – AS unit 1
Motion, Energy and Matter



TUESDAY, 14 MAY 2019 – MORNING
1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	10	
3.	10	
4.	9	
5.	19	
6.	10	
7.	14	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

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The assessment of the quality of extended response (QER) will take place in question **6(b)**.



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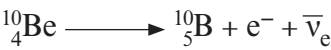
Answer all questions.

1. (a) Mesons, leptons and quarks are three groups of particles. State which of these groups are affected by:

(i) the weak nuclear force; [1]

(ii) the strong nuclear force. [1]

(b) The following interaction shows a β^- decay for a beryllium nucleus.



(i) State the name of the $\bar{\nu}_e$ particle. [1]

(ii) Explain how lepton number and charge are conserved in this interaction. [2]

(c) (i) Write down the number of neutrons in:

I. ${}^{10}_4\text{Be}$ II. ${}^{10}_5\text{B}$ [1]

(ii) On the level of baryons the interaction can be written as:



Show clearly that there is a change in quark flavour in this interaction. [2]

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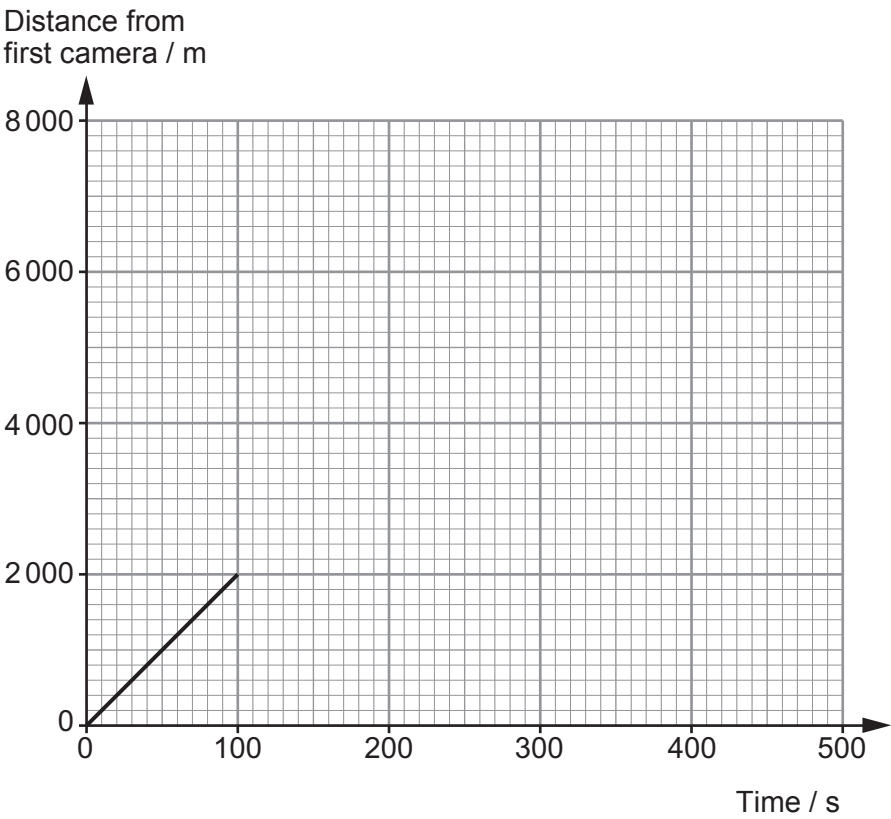
2. (a) State the difference between a vector and a scalar and give one example of each. [2]

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(b) Average (or mean) speed areas are now found on many motorways. A section of motorway has a mean speed of 60 km h^{-1} . The mean speed is monitored between two cameras placed 8 km apart. The graph shows the first 100 seconds of the motion of a car as it travels between the two cameras.



Examiner
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- (i) Show that the instantaneous speed of the car at $t = 50\text{ s}$ is approximately 70 km h^{-1} . [2]

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- (ii) The car covers the 8 km between the two cameras at a mean speed of exactly 60 km h^{-1} . Complete the graph to show this motion. [3]
Space is provided below for your calculations.

- (c) On another stretch of motorway, the speed limit is equivalent to 30 ms^{-1} . The maximum deceleration a normal car can achieve in dry conditions is assumed to be 8 ms^{-2} . Traffic police can estimate the speed of vehicles involved in accidents using the length of the marks made by skidding tyres on the road. In one accident, the skid marks were found to be 85 m long. Investigate whether or not the car that created the skid marks was travelling below the speed limit prior to the accident. [3]

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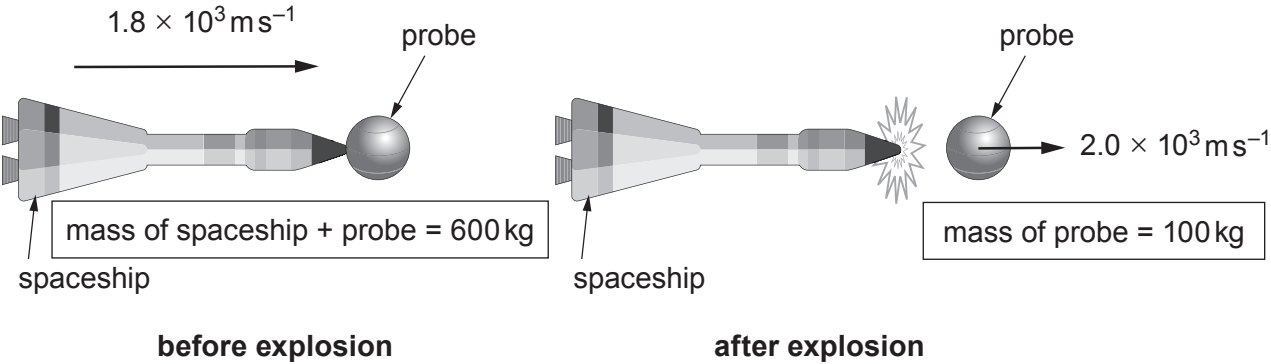
3. (a) State the principle of conservation of momentum. [2]

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(b) A space probe is attached to a large spaceship as shown. The probe can be ejected from the space ship by an explosion. The spaceship and probe together have a mass of 600 kg and they travel in a straight line through deep space at $1.8 \times 10^3 \text{ ms}^{-1}$. Explosives are detonated, separating the probe from the spaceship. Immediately after the explosion the probe, of mass 100 kg, continues in the original straight line at $2.0 \times 10^3 \text{ ms}^{-1}$.



(i) Calculate the velocity of the spaceship immediately after the explosion. [2]

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(ii) Show that the kinetic energy possessed by the probe and spaceship after separation is **greater** than the kinetic energy they possessed before separation. [3]

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Examiner
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(iii) Account for this increase in kinetic energy. [1]

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(c) During the explosion, a mean force, F , acts on the probe for 2.0 ms. Calculate the value of F . [2]

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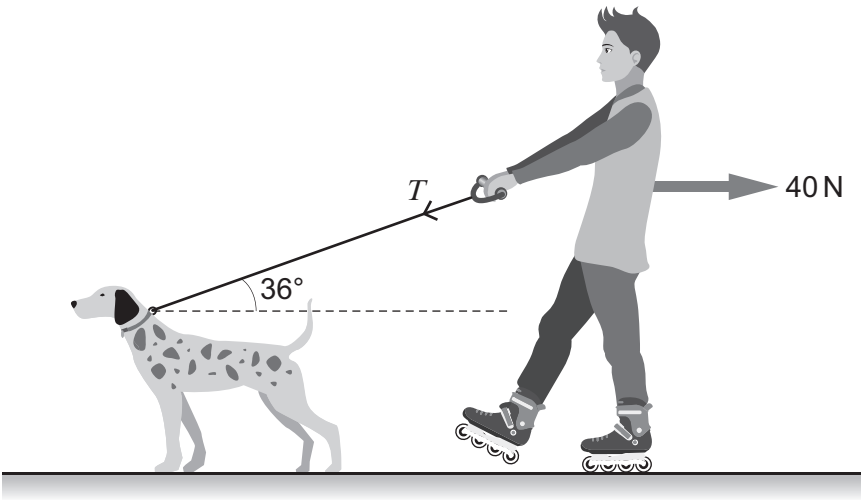
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4. (a) The diagram shows a roller-skater being pulled along a horizontal road at **constant velocity**. The total resistive force to the skater's motion is 40 N.



Show that the tension, T , in the dog lead is approximately 50 N. [2]

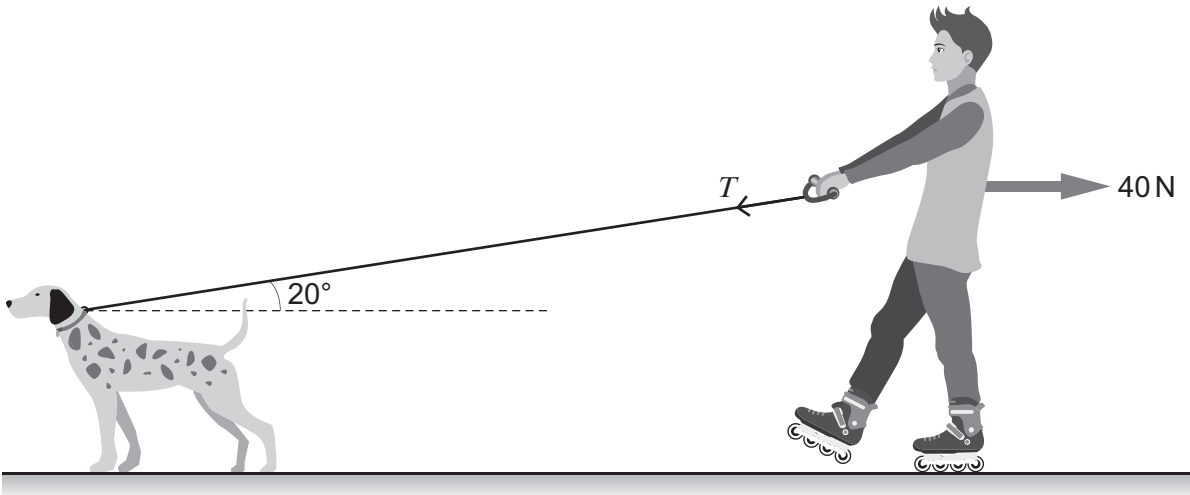
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- (b) Later, the roller-skater, moving at the same velocity as in part (a), extends the dog lead as shown in the diagram below.



- (i) Assuming that the tension, T , is the same as in part (a), explain why the roller-skater initially accelerates. [1]

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Examiner
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(ii) Calculate this initial acceleration, given that the mass of the roller-skater is 35 kg. [3]

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(c) The roller-skater believes that the dog’s rate of doing work is the same, regardless of the length of the dog lead. Use information from (a) and (b) to investigate whether or not this claim is correct. [3]

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5. Sam carries out an experiment to determine the mass of a metal block using the principle of moments.

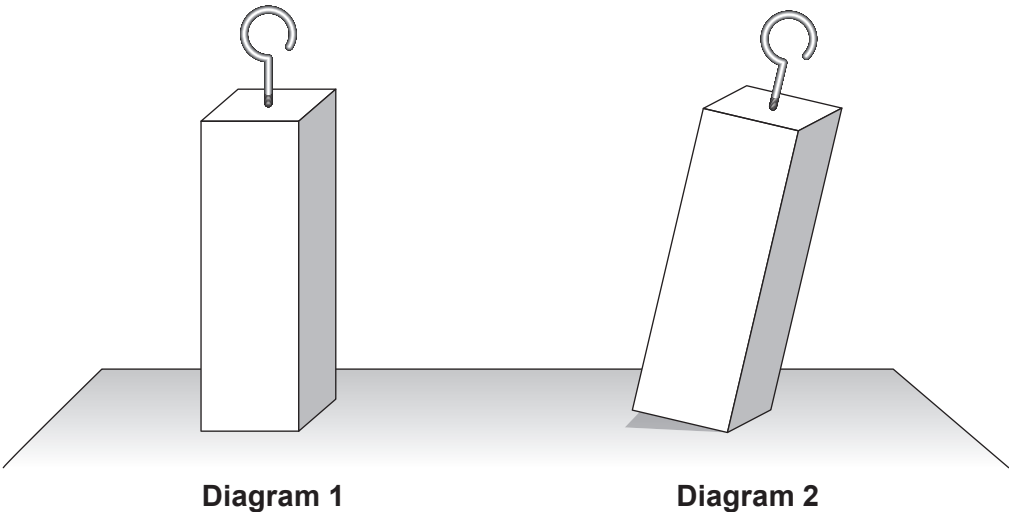
(a) State the principle of moments. [2]

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(b) The metal block is in the form of a cuboid as shown in **Diagram 1**. When setting up the experiment Sam accidentally knocks the metal block so that it tilts to the position shown in **Diagram 2**. The block then returns to the upright position as in **Diagram 1**.



Explain, in terms of centre of gravity, why the block returns to the upright position. [2]

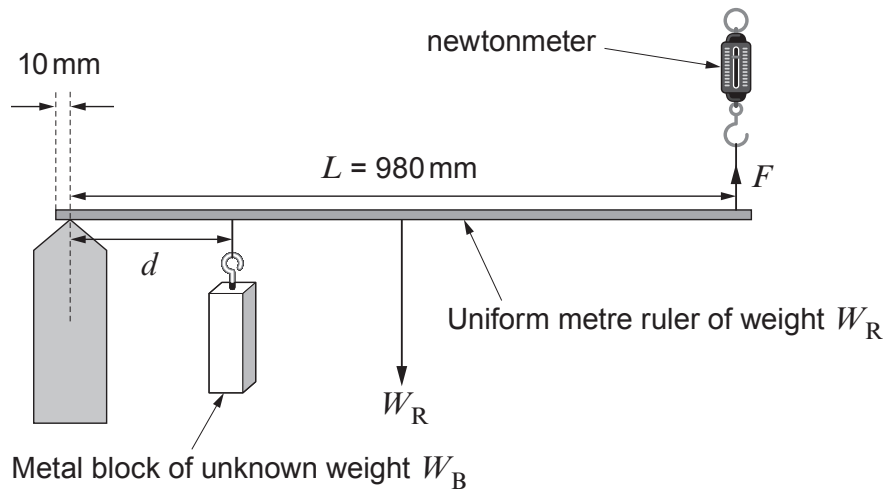
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(c) Sam sets up the experiment as shown using a metre ruler.



Sam uses the newtonmeter to measure the force, F , needed to make the metre ruler horizontal for varying values of d . He carries out the experiment twice and determines a mean value for F for each of the distances d . Sam judges that the ruler is horizontal by eye before taking readings for F . The measurements are recorded in the table below.

Distance, d / mm	Force, F / N		
	Trial 1	Trial 2	Mean
100	2.8	2.6	
250	5.2	5.7	
400	8.8	8.5	
550	11.2	11.6	
700	14.6	14.5	
850	17.2	17.6	

- (i) Complete the table. [1]
- (ii) The following relationship is used to find the weight, W_B , of the block:

$$F = \frac{W_B d}{L} + \frac{W_R}{2}$$

By taking moments about the pivot, show how this relationship is obtained. [2]

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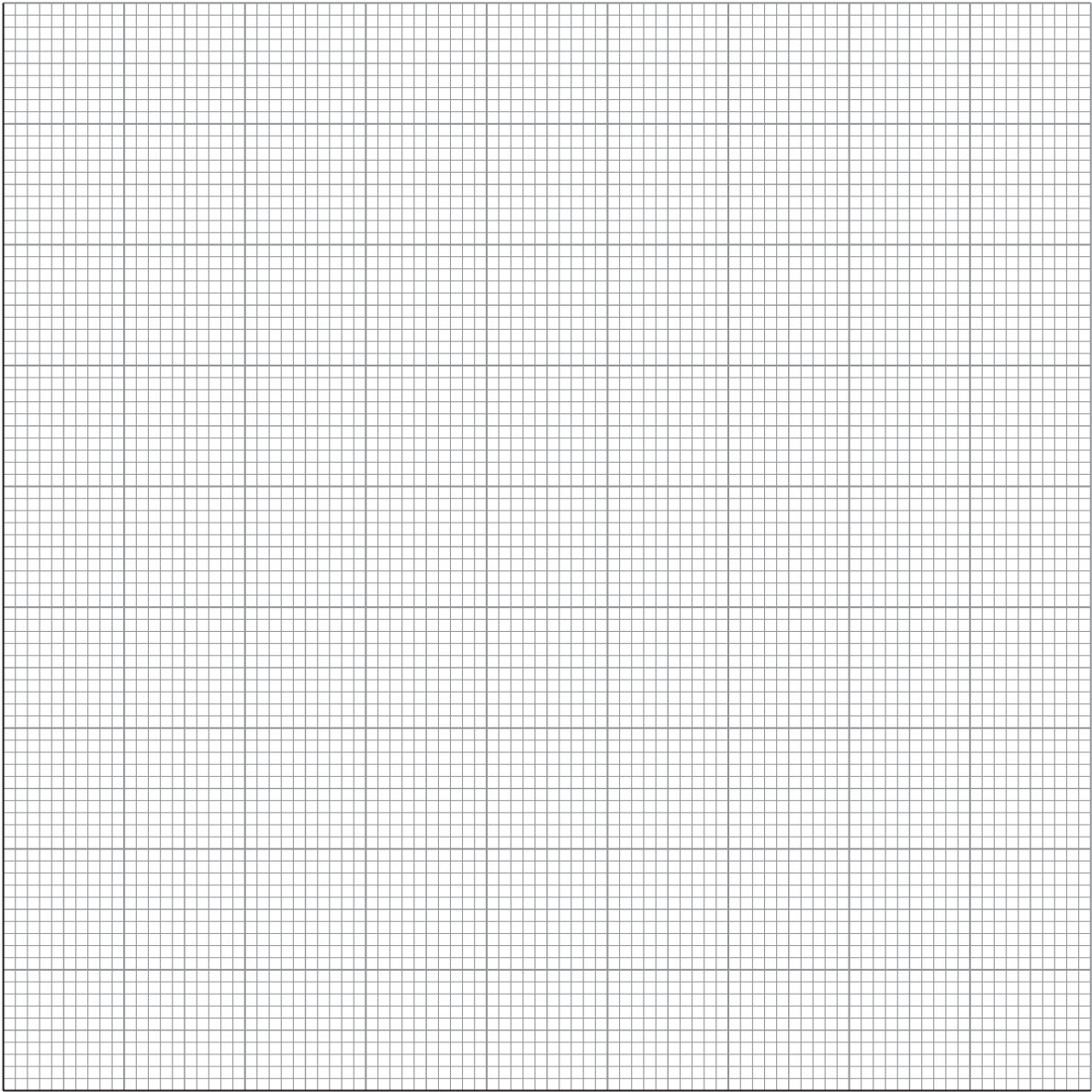
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Examiner
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- (iii) Use the grid to plot a graph of mean F (vertical axis) against d (in mm, on the horizontal axis) and draw a line of best fit. **Error bars should not be included.** [4]



Examiner
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(iv) Use your graph to determine:

I. W_B , the weight of the metal block; [2]

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II. W_R , the weight of the ruler. [2]

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(d) The distances, d , were measured to a resolution of ± 1 mm. Use data from the table of results to show that the percentage uncertainty in mean F is greater than the percentage uncertainty in d , when $d = 100$ mm. [3]

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(e) Suggest **one** change to his method which would improve the quality of Sam's measurements. [1]

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Examiner
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6. (a) Materials can be classified as *crystalline*, *amorphous* or *polymeric*. Explain the meaning of **two** of the terms in italics in terms of microscopic structure. Give **one** example of each of your chosen materials. [4]

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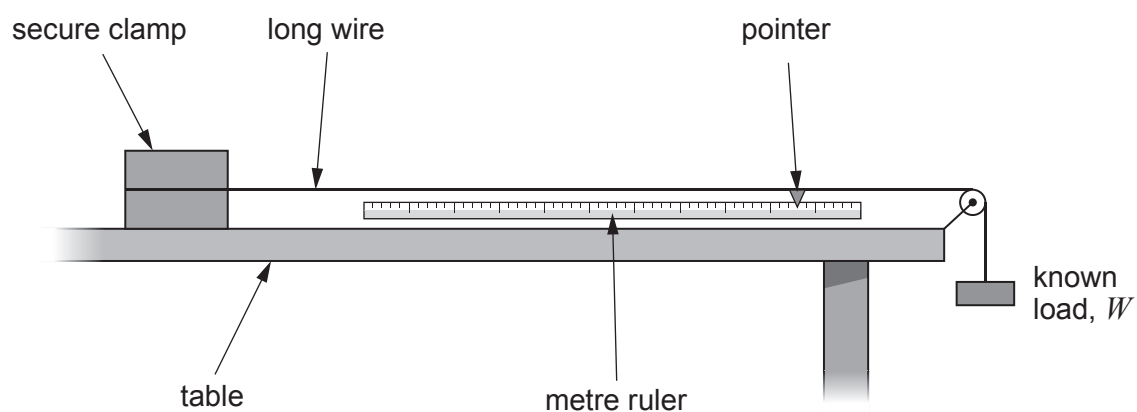
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Examiner
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- (b) The diagram shows apparatus that can be used to determine the Young modulus of a metal in the form of a wire. Describe the measurements which must be made, how they should be used to determine the Young modulus **and** precautions to minimise uncertainties. [6 QER]



Examiner
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7. (a) Describe the main features of the spectrum of a star and state where in the star they arise. [2]

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(b) The table gives some information about two stars.

Star	Luminosity / W	Distance from Earth / m
Sirius	9.7×10^{27}	8.1×10^{16}
Vega	1.5×10^{28}	2.4×10^{17}

(i) Determine the ratio:

$$\frac{\text{Intensity of radiation reaching Earth from Sirius}}{\text{Intensity of radiation reaching Earth from Vega}}$$

[3]

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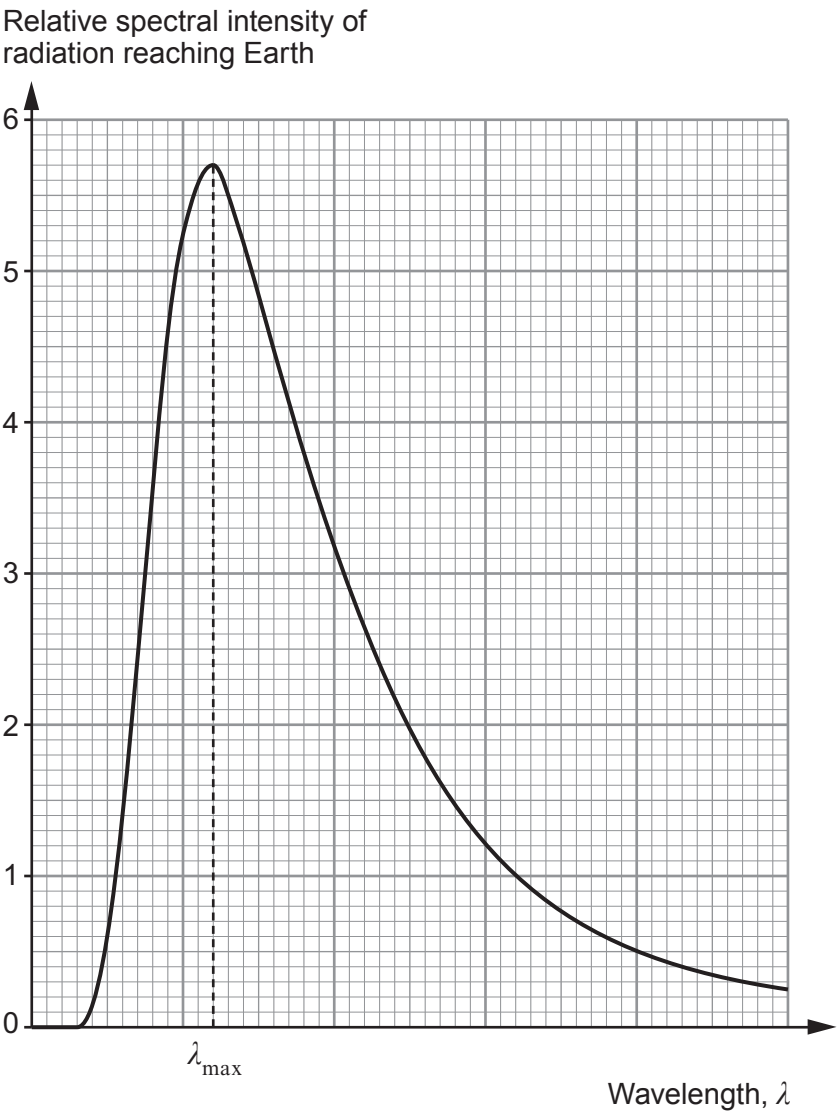
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- (ii) The two stars have similar surface temperatures. Below is a black body radiation curve for Sirius. Using your answer to (b)(i), sketch the expected black body curve for Vega on the same axes. [2]



- (iii) Determine λ_{max} , given that the surface area of Sirius is $1.8 \times 10^{19} \text{ m}^2$. [4]

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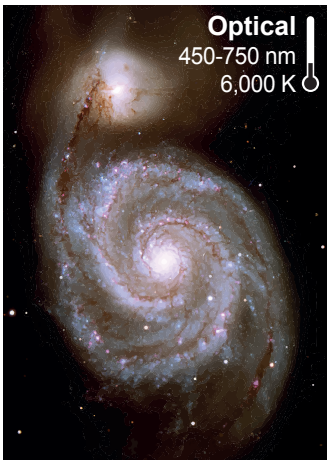
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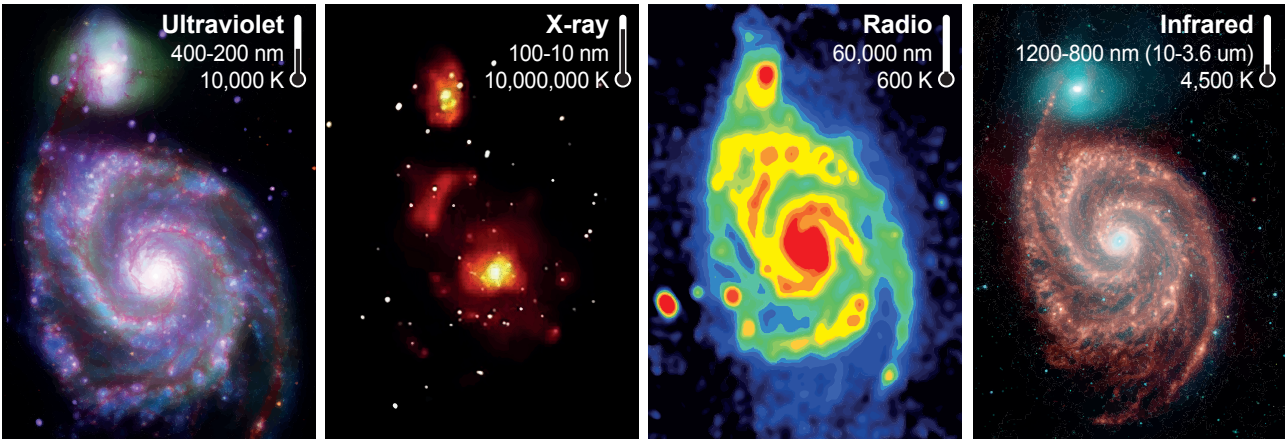


Examiner
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(c) The image below is of the whirlpool galaxy, M51 (or NGC 5194). This is one of the first galaxies to be photographed by astronomers.



Subsequent images of the **same galaxy** are shown below.



Describe how these developments in observational astronomy have advanced the study of the whirlpool galaxy. [3]

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END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		2



GCE AS/A LEVEL

2420U10-1



WEDNESDAY, 18 MAY 2022 – MORNING

PHYSICS – AS unit 1
Motion, Energy and Matter
1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	10	
3.	17	
4.	14	
5.	9	
6.	12	
7.	9	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The total number of marks available for this paper is 80.
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **5(a)**.



JUN222420U10101

Answer **all** questions.

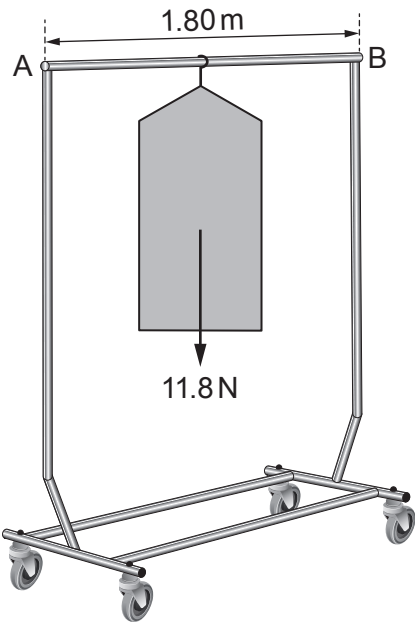
1. (a) State the conditions needed for a body to remain in equilibrium. [2]

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(b) The diagram shows a simple clothes carrier. It consists of a uniform horizontal beam (AB) of length 1.80 m resting on two vertical rods.



When the clothes carrier supports a coat weighing 11.8 N **at its centre**, a force of 8.0 N is exerted on each rod. Show that the weight, W , of the horizontal beam is approximately 4 N. [2]

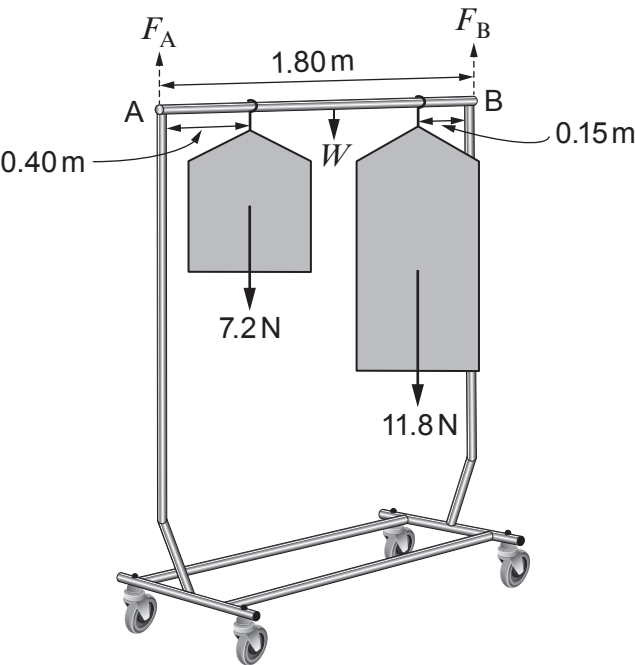
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- (c) The coat is now moved to a new position and a jacket weighing 7.2 N is hung from the beam, all of which is shown below.



Calculate forces F_A and F_B .

[5]

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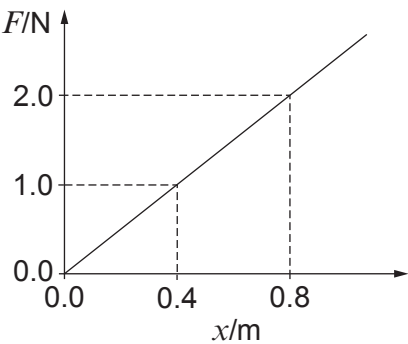
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2. (a) The graph shows how the force, F , needed to stretch a simple spring varies with the extension, x , of the spring. The spring obeys Hooke's law.



- (i) Calculate the work done in stretching the spring 0.4 m from its unstretched position.

[2]

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- (ii) Explain why the work done to stretch the spring 0.8 m from its unstretched position is **not twice** your answer to part (a)(i).

[2]

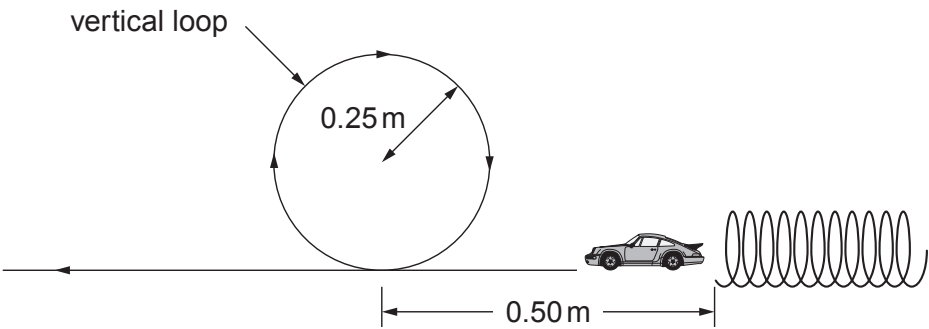
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- (b) The spring in part (a) is used to propel a toy car of mass 30 g along a track which contains a vertical loop of radius 0.25 m.



Examiner
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- (i) The spring is compressed by 0.4 m. The car is placed against the end of the spring, which is then released. Assuming that all of the spring's energy is transferred to the car and that no resistive forces act on it, calculate the speed of the car at the top of the loop. [3]

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- (ii) In reality resistive forces act on the car to the extent that the car loses 5% of its initial energy to frictional forces in moving from the spring to the top of the loop. Calculate the mean resistive force acting on the car during this motion. [3]

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3. Drones such as the one shown in the picture have an increasing number of uses in society. This question considers some of the Physics associated with drones.



(a) (i) State Newton’s third law of motion.

[1]

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(ii) The drone in the picture has four rotors and is shown hovering stationary in the air. The table describes two vertical forces acting on the drone. According to Newton’s third law each of these forces is part of a pair of forces. Identify the other force in each case.

[2]

Forces acting vertically downwards	Forces acting vertically upwards
Gravitational force of Earth on drone	
.....	Force of air on rotors

(b) The drone is able to hover when each rotor produces a downward flow of air of speed 22 m s^{-1} . The momentum given to the air by the 4 rotors produces an upward force which keeps the drone stationary in the air.

(i) State the relationship between force and momentum.

[1]

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- (ii) The rate of change of momentum of the air **from the 4 rotors** can be expressed as:

$$4\pi r^2 \rho v^2$$

where r is the radius of each of the rotor blades, v is the speed of the air which is set in motion and ρ is the density of the air. Use this expression to show that the weight of the drone is approximately 20 N. [$\rho = 1.3 \text{ kg m}^{-3}$, $r = 5.0 \times 10^{-2} \text{ m}$] [2]

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- (iii) Calculate the vertical acceleration of the drone when v is increased to 24 ms^{-1} . [4]

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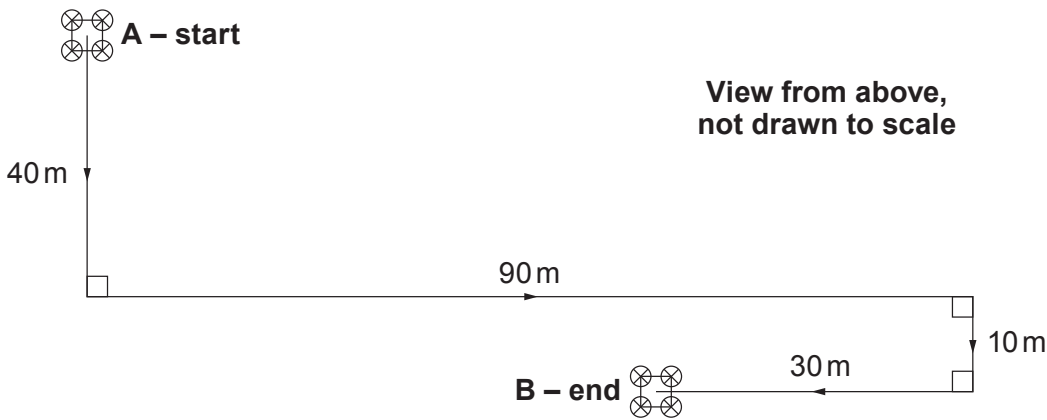
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(c) (i) Define *velocity*. [1]

(ii) Bill flies the drone from **A** to **B** in a time of 32 s along the path shown below. The drone is kept at the same height above the ground for the entire flight.



Bill's friend Ted believes that, in flying this path (from **A** to **B**), the mean speed of the drone is more than twice the magnitude of its mean velocity. Bill disagrees. Use the information given to determine who is correct. [4]

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(d) Drones have many applications in today's society. Give **one** benefit and **one** risk of drone technology. [2]

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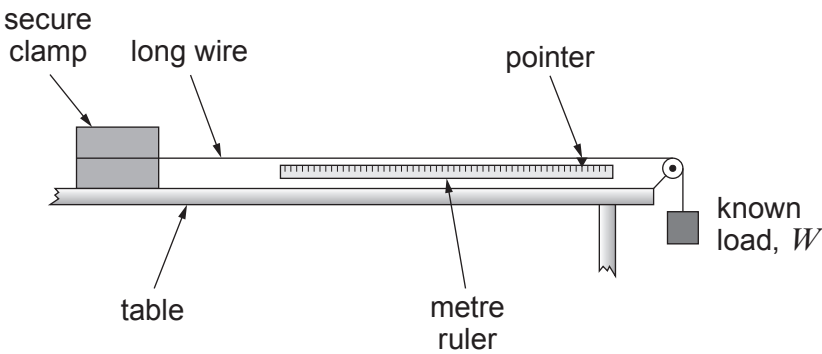
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4. Karen is an A level Physics student who has been absent for some of her Physics practical lessons. During her absence, the other students in the class have been using the apparatus below to identify the metal in a wire by finding the Young modulus of the metal and comparing it with known values.



In order to catch up with the other students, Karen decides to use the same apparatus to carry out a quick experiment to find the Young modulus. Her notes are shown in the box below.

- Take single reading of diameter using a micrometer of resolution 0.01 mm.
Diameter = 0.18 mm
- Use a metre ruler of resolution 1 mm to measure the initial length of the wire.
Initial length = 1.940 m
- Set up the apparatus and add a **mass of 1.0 kg** to the end of the wire.
Measure extension.
Extension = 6.0 ± 0.5 mm

(a) Use Karen's notes to answer the following questions.

- (i) Calculate the cross-sectional area of the wire in m^2 and show that the **percentage** uncertainty in its value is approximately 11%. [2]

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- (ii) Show, with an appropriate calculation, that the uncertainty in the length reading can be considered negligible. [1]

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Examiner
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(b) (i) Determine the Young modulus of the metal. [2]

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(ii) Karen is told by her teacher that the percentage uncertainty in the load is 5%. Use this information along with Karen’s notes and your answers to (a) to calculate the **absolute** uncertainty in the Young modulus. Give your answer in GPa and to an appropriate number of significant figures. [4]

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(c) The following table shows Young modulus values of the wires available to the students.

Metal	Young modulus / GPa
aluminium	70
zinc	100
bronze	103
copper	117
nickel	166

Karen’s friend Jack determined the Young modulus of the metal to be 126 ± 20 GPa. They both use the above table to determine the metal in their wires. Discuss how Jack’s conclusions differ from Karen’s. [2]

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Examiner
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(d) Jack had more time to carry out this experiment than Karen. Suggest differences in the way Jack carried out the experiment (i.e. procedure) **and** in the way that he analysed his results which led to him reaching a different conclusion to Karen. [3]

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Examiner
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5. (a) **Hadrons** are a group of particles. Write a detailed account of hadrons, including how they may be subdivided into other groups of particles, giving examples. [6 QER]

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(b) A high energy interaction between a proton and a pion is shown below.

$$p + \pi^- \rightarrow n + \pi^+ + \pi^-$$

Use conservation laws to show that the interaction is possible. [3]

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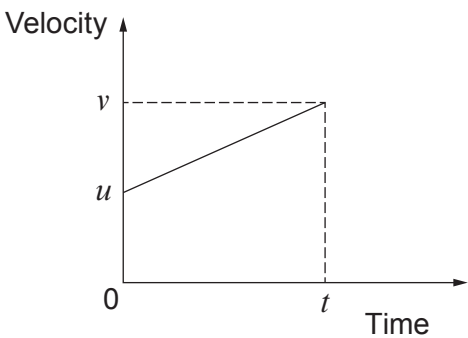
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6. (a) A velocity-time graph is given for a body which is accelerating uniformly for a time, t .



Using the graph, show clearly that: [3]

$$x = ut + \frac{1}{2}at^2$$

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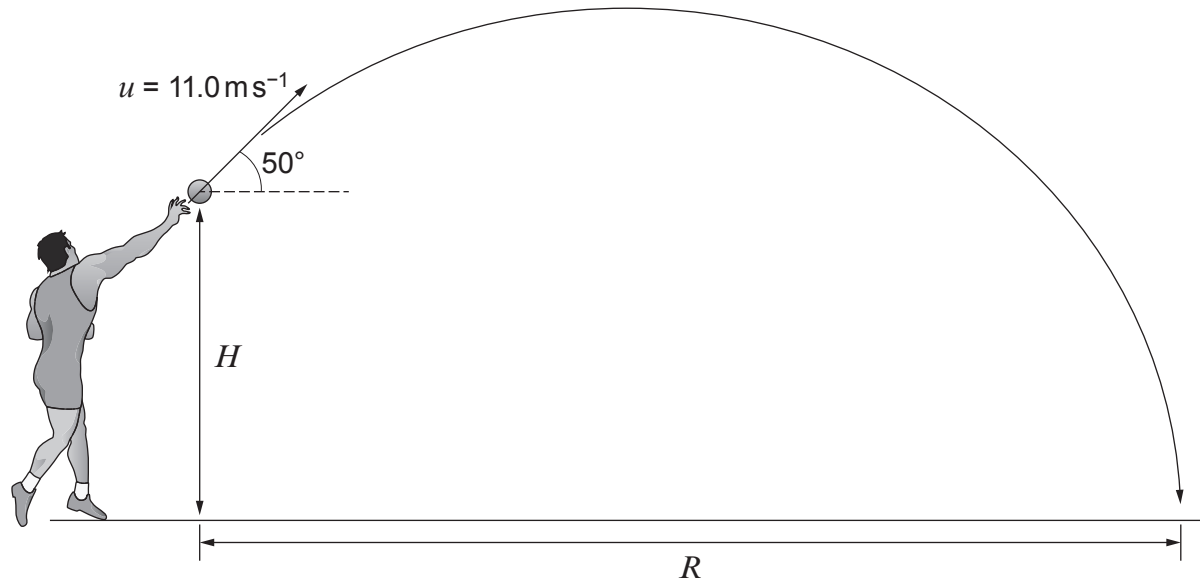
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(b) The shot put is an event in athletics. The shot (a heavy metal ball) is thrown from a standing position so that it lands on the ground a horizontal distance, R , from the thrower.



Examiner
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Steve is a shot putter. He throws a shot with a velocity of 11.0 m s^{-1} in a direction 50° to the horizontal. The shot takes 1.95 s to reach the ground.

- (i) Show that the vertical component of the initial velocity is approximately 8 m s^{-1} . [1]

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- (ii) Hence calculate, H , the vertical distance above the ground from which the shot was thrown. [3]

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- (iii) Calculate the horizontal distance, R , for this throw. [2]

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- (c) Steve makes the following comment to Bryn, a fellow competitor.

“Throwing the shot at an angle even greater than 50° to the horizontal will increase the horizontal distance travelled by the shot because the time of flight will increase.”

Consider whether or not it would be a good idea for Bryn to follow Steve’s advice. Justify your answer. [3]

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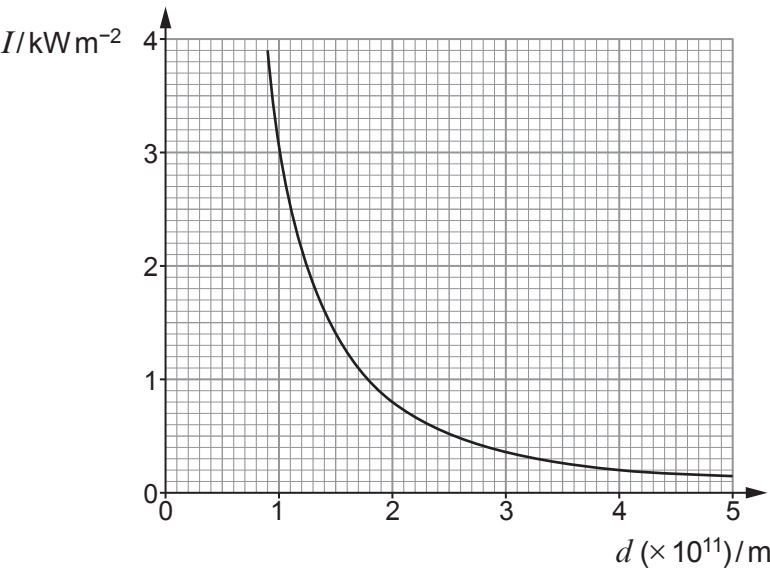


7. (a) Stars are very good approximations of *black bodies*. Explain what is meant by the term *black body*. [1]

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(b) The graph shows how the intensity of electromagnetic radiation, I , from the Sun varies with the distance from its centre, d .



(i) Confirm that the graph shows an inverse square relationship between I and d . [2]

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(ii) Show that the Sun's luminosity is approximately 4×10^{26} W. [2]

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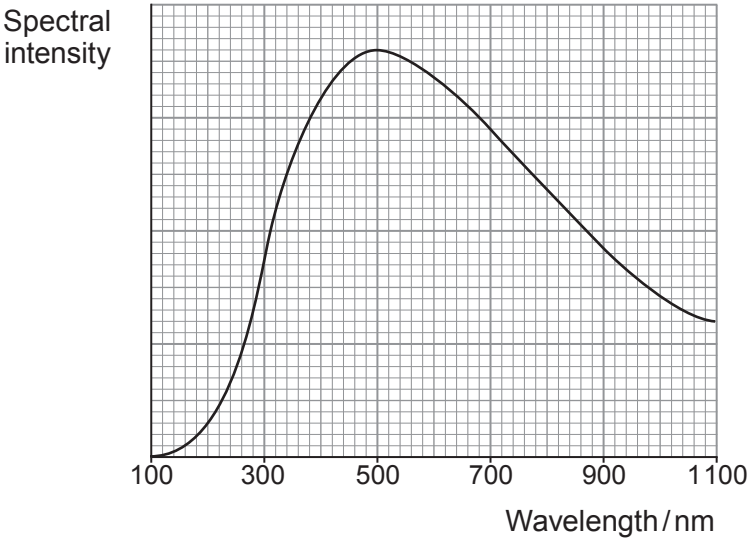
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- (c) The graph shows how the intensity of the radiation incident on the Earth from the Sun is distributed across the spectrum.



Use information from this graph, along with your answer to (b)(ii), to calculate the radius of the Sun. [4]

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END OF PAPER

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Surname	Centre Number	Candidate Number
First name(s)		2



GCE AS/A LEVEL

2420U10-1



THURSDAY, 18 MAY 2023 – AFTERNOON

PHYSICS – AS unit 1
Motion, Energy and Matter

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	11	
3.	15	
4.	6	
5.	11	
6.	11	
7.	10	
8.	9	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the additional page(s) at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The total number of marks available for this paper is 80.
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question 4.



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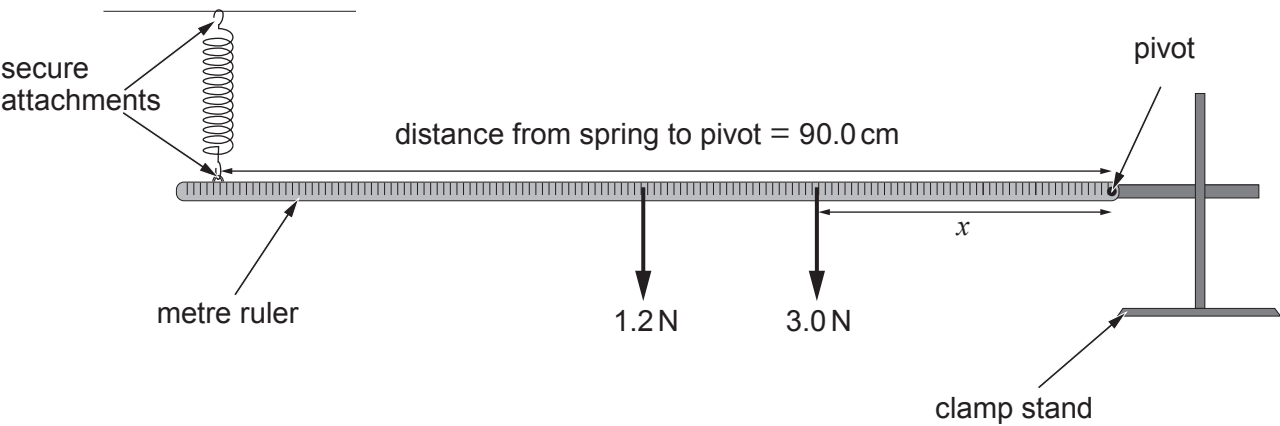
Answer **all** questions.

1. (a) Define the spring constant, k . [1]

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(b) The following apparatus is set up to investigate moments. A horizontal uniform metre ruler weighing 1.2 N is freely pivoted at one end. The ruler is suspended by a spring of spring constant, $k = 20\text{ N m}^{-1}$, at a point 90.0 cm from the pivot, and a load of 3.0 N is suspended at a distance, x , from the pivot.



Use the following information to calculate x .

Original (unstretched) length of spring = 10.0 cm
Stretched length of spring when the ruler is horizontal = 17.0 cm [4]

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(c) Two springs of the same type used in part (b) are now connected in series as shown.



Laura believes that connecting the springs in this way will result in the overall spring constant being 10 N m^{-1} . Aled believes that the overall spring constant will be 40 N m^{-1} . Explaining your answer determine who, if either, is correct. [2]

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2. Tomos and Jerry wish to determine the material from which a ball bearing is made. They both decide to determine the density of the metal of the ball bearing. However, they choose different methods.

Tomos' method:

Tomos uses digital callipers to measure the diameter of the ball bearing. He then uses a digital balance to determine the mass of the ball bearing. He obtains the following values:

Diameter of ball bearing = $18.76 \pm 0.02 \text{ mm}$

Mass of ball bearing = $26.3 \pm 0.5 \text{ g}$

Jerry's method:

Jerry measures the volume directly by submerging the ball bearing in water in a measuring cylinder. He also uses the same digital balance to determine the mass of the ball bearing and obtains the following values:

Volume of ball bearing = $3.4 \pm 0.1 \text{ cm}^3$

Mass of ball bearing = $26.3 \pm 0.5 \text{ g}$

(a) Using Tomos' values:

- (i) Calculate the volume of the ball bearing in cm^3 and show that its **percentage** uncertainty is approximately 0.3%. [3]

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- (ii) Hence calculate the density of the ball bearing along with its **absolute** uncertainty, giving your answer to an appropriate number of significant figures. [4]

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- (b) Jerry believes that, since he is measuring the volume directly, the absolute uncertainty in his value of density will be less than that found by Tomos. Determine whether or not he is correct. [2]

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- (c) The table gives the density of some common metals and alloys.

Metal	Density/g cm ⁻³
tin	7.3
stainless steel	7.5
iron	7.9
brass	8.3

Tomos and Jerry use this information to determine the metal from which the ball bearing is made. Explain how Tomos' conclusion differs from Jerry's. [2]

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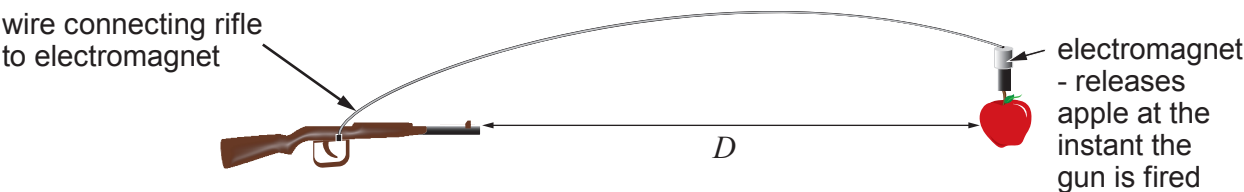
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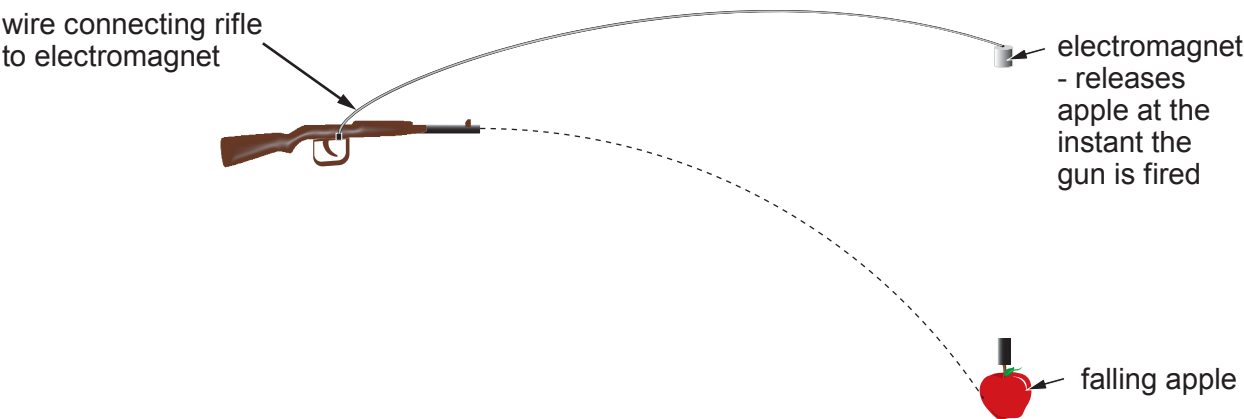


3. (a) The following demonstration is sometimes used to explain projectile motion. In the version shown, a toy rifle is situated a horizontal distance, D , from an apple suspended at a large height above the ground.

Initial set-up



Initially, the rifle is aimed horizontally at the apple. **At the instant** the gun is fired an electromagnet releases the apple so that it falls vertically. The path of the pellet from the gun is shown below.



- (i) Describe and explain the motion of the pellet from the instant the gun is fired. [Ignore the effects of air resistance.] [2]

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- (ii) The distance D is now increased and the experiment repeated. Describe one difference **and** one similarity that an observer viewing the apple would see in this case. [2]

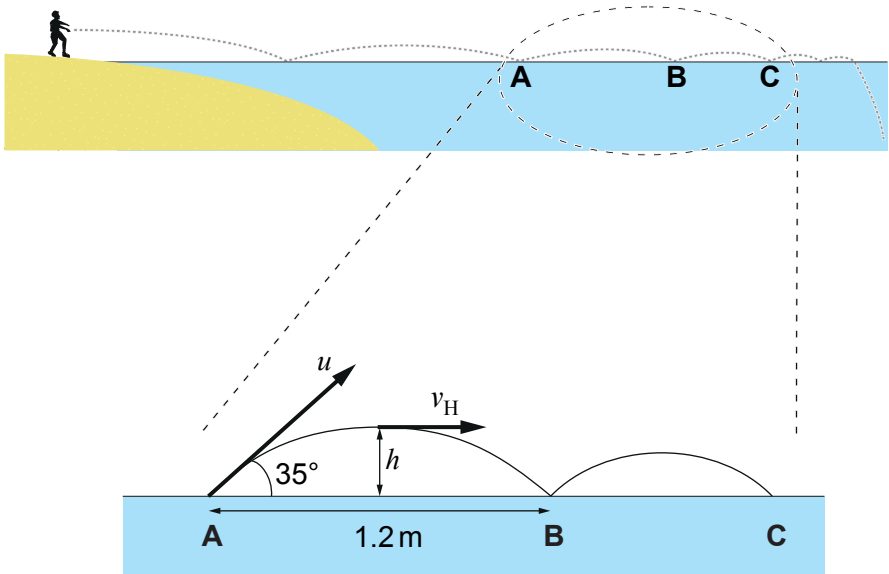
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- (b) The diagram shows a person ‘skimming’ a stone across the surface of a still pond. A magnified view of part of the stone’s motion is also shown. [Ignore the effects of air resistance.]



- (i) Calculate v_H , the horizontal component of the velocity, u , given that the time taken for the stone to move from **A** to **B** is 0.40 s. [1]

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- (ii) Calculate h , the maximum height achieved by the stone between **A** and **B**. [4]

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(iii) Calculate the **total** energy of the stone at height, h , given that it has a mass of 0.10 kg. [3]

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(c) The thrower believes that the stone loses 20 % of its energy every time it impacts with the water. Investigate whether or not this is true **for the impact at B**, given the fact that the stone’s velocity just before impacting the water **at C** is 3.3 m s^{-1} . [3]

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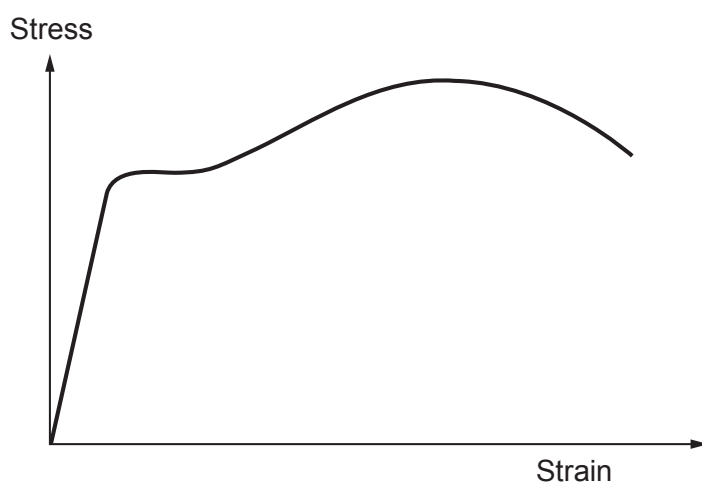
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4. A stress against strain graph for a ductile metal is shown.



Identify and describe the key features **shown by this graph**. Reference to atoms, ions or molecules is **not required**. [6 QER]



5. (a) Newton’s second law of motion is sometimes expressed as $\Sigma F = ma$. Explain the term ΣF , giving an example to illustrate your answer. [2]

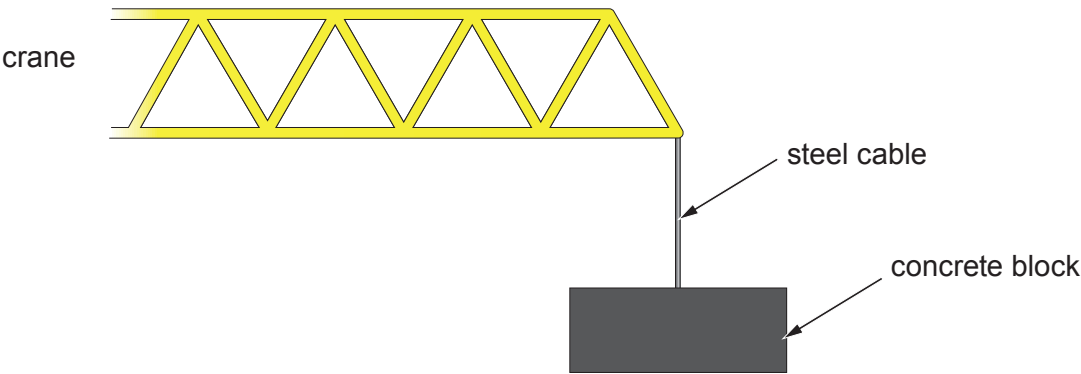
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- (b) A crane uses a steel cable to lift heavy objects on a building site. The crane operator is told that the maximum safe lifting force of the crane is 16 000 N.



- (i) A concrete block of mass 1 500 kg is attached to the crane. Calculate the maximum safe upward acceleration of the block. [3]

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- (ii) The crane operator assumes that the crane is safe to lift any load up to 16 000 N. Discuss whether or not he is correct. [2]

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- (c) The steel cable has a cross-sectional area of $2.0 \times 10^{-3} \text{ m}^2$ and a Young modulus of $2.0 \times 10^{11} \text{ N m}^{-2}$. As the concrete block moves **upwards** the tension in the cable changes depending on whether the block is accelerating, decelerating or moving at constant speed. At one point in its motion the strain in the cable is 3.2×10^{-5} . Describe the motion of the block at this point. Explain your answer. [4]

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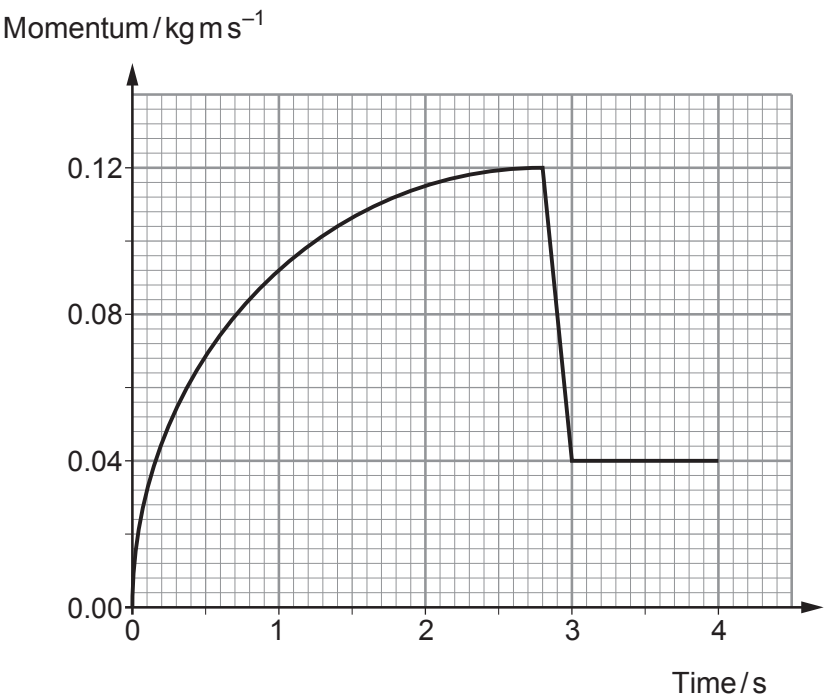
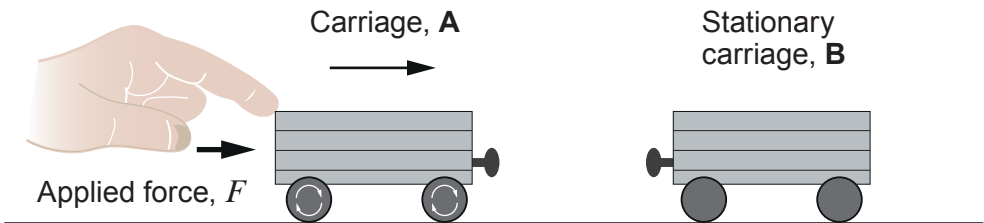
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6. A toy train carriage, **A**, is accelerated from rest by an applied force in a straight line on a smooth track towards a stationary carriage, **B**. A graph of momentum against time for carriage **A** is shown below.



- (a) Estimate, from the graph, the resultant accelerating **force** acting on carriage A at a time of 1.0 s. [3]

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- (b) Impact between carriage A and carriage B occurs between 2.8 s and 3.0 s. Just before impact the applied force is removed. State how the graph confirms that the applied force is removed, and explain why its removal enables you to calculate the momentum given to carriage B. [3]

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- (c) (i) **Draw, on the graph opposite,** a line showing the momentum of **carriage B** between 0 s and 4 s. [3]

- (ii) Hence calculate the speed of **carriage B** after impact given that its mass is 0.16 kg. [2]

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7. (a) In the table below, statements are given which describe different particles or interactions.
Complete the table by naming the particle or interaction described. [4]

Description of particle or type of interaction	Name of particle or interaction
The quark combination of this particle is uud .	
The electron and electron neutrino are examples of this group of particles.	
Antibaryons are a combination of three of these.	
Neutrino involvement and quark flavour changes are exclusive to this type of interaction.	

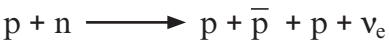
- (b) An antiparticle has a quark composition of $\bar{u}\bar{d}\bar{d}$. Determine its charge and identify the particle. Show your working clearly. [2]

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- (c) (i) Consider the following hypothetical interaction.



The reaction is **not possible** because it does not obey one or more of the conservation laws. By considering baryon number, lepton number and charge, show which law(s) are obeyed and which are not. [3]

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- (ii) Jon suggests that replacing the electron neutrino (ν_e) with a pi-zero pion (π^0) would allow the reaction to occur. Explain whether or not he is correct. [1]

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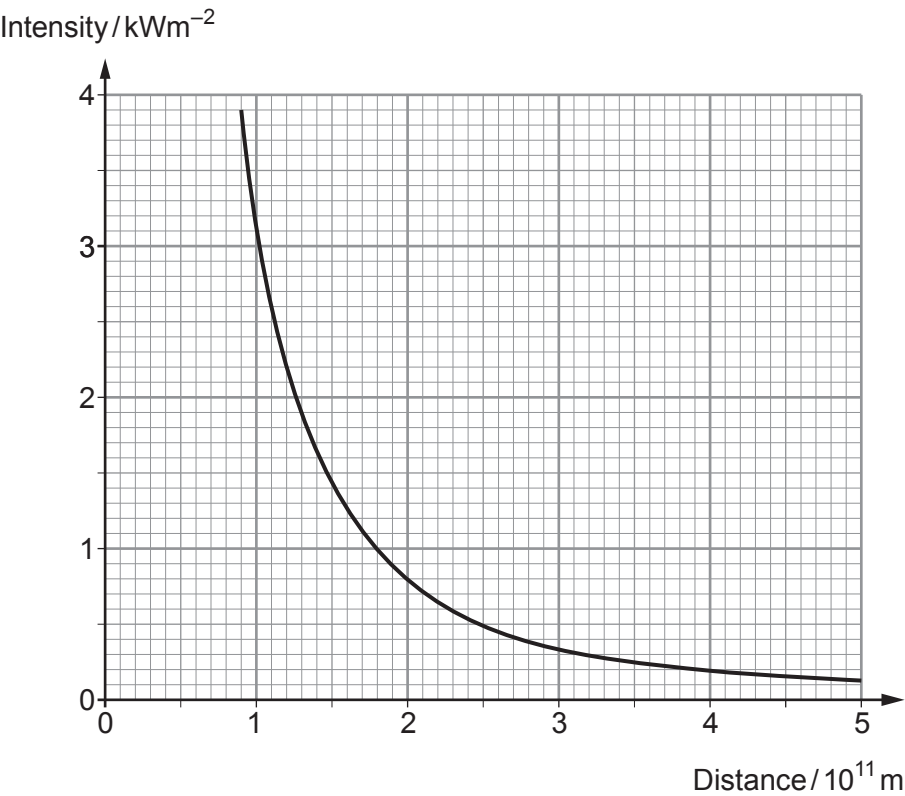
8. (a) The solar spectrum includes an absorption spectrum. Describe the appearance of this spectrum **and** state where in the Sun the absorption occurs. [2]

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- (b) The graph shows how, outside the Sun, the intensity of electromagnetic radiation from the Sun varies with the distance from its centre.



Use information from the graph to determine the total power emitted by the Sun. [3]

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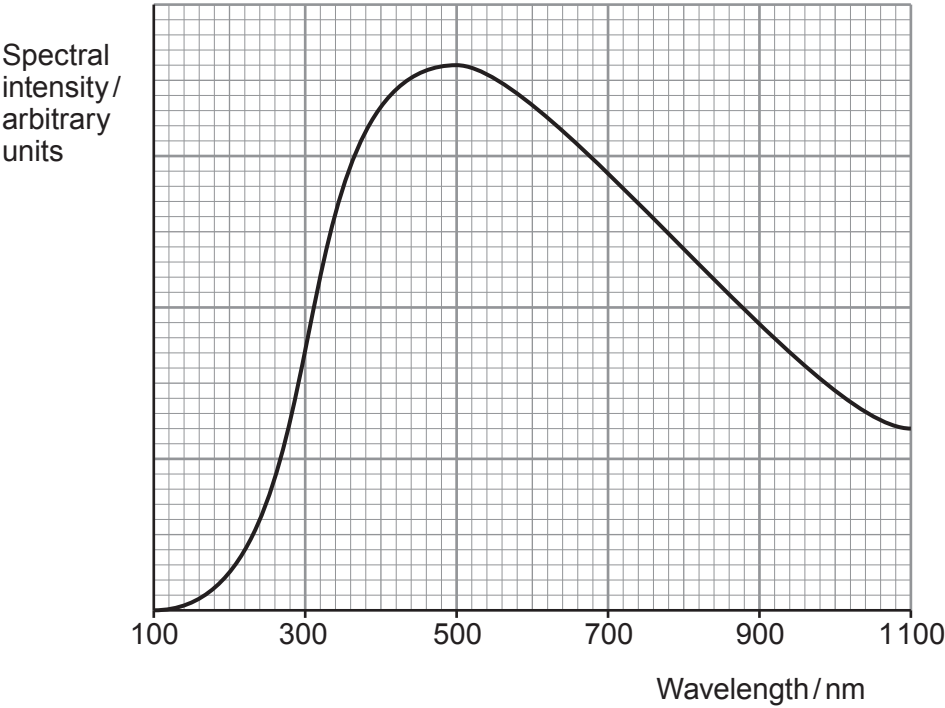
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- (c) The diagram below shows how the intensity of the radiation incident on the Earth from the Sun is distributed across the spectrum.



Determine whether or not the answer you obtained in part (b) and information which can be obtained from the above spectrum are consistent with each other.
[Surface area of the Sun = $6.2 \times 10^{18} \text{ m}^2$]

[4]

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END OF PAPER

